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Physics-Informed Multi-Target RF Field Reconstruction over Heterogeneous Terrain Graphs under Spatially Blocked Evaluation

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ABSTRACT Empirical models like FSPL, Okumura–Hata, and COST 231 lose accuracy where terrain, vegetation, and urban fabric shape sub-kilometer propagation beyond coarse categories, and the evaluation partition changes what a comparison supports. GNN-RF, a heterogeneous GATv2 graph neural network, is trained on 12.96×10^6-node terrain graphs from four São Paulo cities to reconstruct path loss, RSSI, and coverage, with a physics-constrained decoder and four loss regularizers: free-space floor, monotonicity, variance, and NDVI sign. Under spatial blocking (80 runs), GNN-RF's all-node RSSI MAE is $0.395\,\mathrm{dB}$ (95% CI $[0.309, 0.480]$), on average $0.746\,\mathrm{dB}$ below each cell's best calibrated empirical model (paired $p{=}2.59 \times 10^{-4}$), winning 15 of 16 cells. Over the 17.04% valid-target subset, its MAE is $2.012\,\mathrm{dB}$ (cell mean, Lins Q1 included), against $4.362\,\mathrm{dB}$ for the best calibrated empirical model of each cell on those nodes, ahead in 12 of 16 cells. GNN-RF's all-node MAE is $0.247\,\mathrm{dB}$ below a graph-free control matched in nominal parameters (city-level interval crossing zero, $p{=}0.205$); on valid-target nodes, GNN-RF and a node-wise RadioUNet adaptation with only antenna-to-terrain graph relations are 0.727 and $1.506\,\mathrm{dB}$ below that control, respectively, and their difference, like the effect of retraining without penalties, is inconclusive relative to the pre-specified margin. Uncertainty quantification fails a pre-specified joint coverage criterion: MC Dropout qualifies in 3 of 16 cells, split-conformal in 1, and a residual-based baseline in 2. Labels come from a deterministic propagation generator, so errors measure fidelity to simulated references, not physical-world prediction error.

INDEX TERMS graph neural networks, heterogeneous graphs, physics-informed learning, RF propagation modeling, topography, transfer learning, uncertainty quantification

I. INTRODUCTION

SUB-KILOMETER radio frequency (RF) propagation over morphologically heterogeneous terrain challenges the assumptions of classical coverage models. Free-Space Path Loss (FSPL), Okumura–Hata [1], and COST 231 [2] estimate attenuation through closed-form expressions derived from large-scale measurements, but reduce local relief, vegetation, and urban structure to a limited set of terrain categories [3]. Local ground cover causes location variability over areas of a few hundred meters [4], shadow fading is spatially correlated [5], and vegetation and buildings cause large-scale fading [6]. Coverage prediction depends on path-loss estimates [3], yet conventional empirical methods cannot provide location-specific, high-fidelity predictions in complex propagation environments [7].

This limitation is especially relevant in Brazil, where the vegetated suburban and rural environments with tropical species differ from the temperate-clime conditions on which most models of propagation through vegetation are based [8], and where Okumura–Hata departs from field measurements of sub-GHz links in the Amazon [9]. Elsewhere, at 900 MHz in Cartagena, Spain, classical empirical models have errors well above those of models conditioned on local geospatial descriptors [10], so expanding sub-GHz coverage into sub-

urban areas calls for site-specific predictions beyond what closed-form models can provide.

Learned radio-map estimators [11]–[13], one of which reports gains over an empirical model [12], still face structural limitations for multi-city deployment (Section II). Grid-based models cannot natively represent irregular terrain sampling; physics priors often stop at a path-loss floor and omit vegetation or terrain-gradient effects; evaluations are commonly limited to a single city or synthetic scenarios; and comparisons across architectures frequently use different feature sets, losses, and optimization schedules. These limitations make it difficult to separate gains due to terrain representation from gains due to training design, especially without identical-treatment comparisons and anti-leakage controls.

GNN-RF addresses these gaps by operating natively on the heterogeneous terrain graph instead of on a rasterized proxy of it. Each city quadrant becomes a graph of 12.96×10^6 nodes with three edge types and 18 topographic and spectral features, encoded by a multi-head GATv2 stack and decoded into a five-channel RF output by a physics-constrained head. The output activations bound each channel, and the loss penalizes departures from free-space bounds, distance–attenuation monotonicity, variance preservation, and NDVI sign consistency.

Training follows a sequential transfer protocol across geographic quadrants and is evaluated over 80 runs under spatially blocked partitions spanning four cities in São Paulo state, complemented by cross-city transfer. Because every target label is produced by a fully specified deterministic propagation generator (detailed in the Supplementary Material) rather than measured in the field, the reported accuracy bounds reconstruction fidelity relative to simulation-derived references, not physical-world prediction error. Evaluation design matters for such a target: on the same reference field, a nearest-neighbor predictor has an error roughly an order of magnitude larger under the spatially blocked split used throughout this work than under a random node split, in every city (Section V-A).

The contributions of this work are fivefold. We design a heterogeneous GNN that pairs a multi-head GATv2 encoder with a physics-constrained decoder that jointly predicts total path loss, RSSI, and coverage, supervising vegetation and terrain-diffraction attenuation against the corresponding components of the reference generator. We formulate a composite physics-informed objective that joins a Huber regression term with four domain regularizers: an FSPL-floor penalty, distance–attenuation monotonicity, variance preservation, and NDVI sign consistency; we test their contribution by retraining the model in two arms of 80 runs, one without any of the four terms and one without the monotonicity term alone. We define a sequential transfer-learning protocol over quadrants (Q1→Q4) and test it against training every quadrant from scratch.

We evaluate GNN-RF under a spatially blocked protocol that limits spatial leakage between training and test [14], a split-related form of data leakage [15], against calibrated empirical models, a graph-free control matched in nominal parameters, edge-relation ablations, and, as a learned baseline trained under the same blocked protocol with the same loss and output decoder, an in-house, node-wise adaptation of RadioUNet [11]. We evaluate Monte Carlo Dropout [16] uncertainty across all 16 city–quadrant pairs at five nominal levels, with split-conformal and residual-based comparisons.

II. THEORETICAL FOUNDATION AND RELATED WORK

Heterogeneous message passing supplies an inductive bias that supports a multi-relation terrain graph, since it assigns separate parameters to each relation type instead of forcing one rule across every interaction. A heterogeneous graph is the tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}, \phi, \psi)$, with node set $\mathcal{V}$, edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, relation types $\mathcal{R}$, and type-assignment maps $\phi : \mathcal{V} \to \mathcal{T}_V$ and $\psi : \mathcal{E} \to \mathcal{T}_E$ [17]. Over such graphs the Graph Attention Network [18] learns adaptive neighbor weights, and GATv2 [19] computes dynamic attention, whereas the original formulation computes only static attention:

$$\alpha_{ij} = \frac{\exp\big(\mathbf{a}^\top \mathrm{LeakyReLU}(\mathbf{W}[\mathbf{h}_i \| \mathbf{h}_j \| \mathbf{e}_{ij}])\big)}{\sum_{k \in \mathcal{N}(i)} \exp\big(\mathbf{a}^\top \mathrm{LeakyReLU}(\mathbf{W}[\mathbf{h}_i \| \mathbf{h}_k \| \mathbf{e}_{ik}])\big)} \tag{1}$$

where $\mathbf{W}$ is a learnable weight matrix, $\mathbf{a}$ the attention vector, $\mathbf{h}_i$ the node embedding, $\mathbf{e}_{ij}$ the edge feature, and $\|$ denotes concatenation; K parallel heads are concatenated before projection, as in GAT [18]. Relation-specific message passing, built on the scheme in which each node sums messages from its neighbors and then updates its state [20], composes these per-relation attention maps into a single node update,

$$\mathbf{h}_i^{(\ell+1)} = \sigma\left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_r(i)} \alpha_{ij}^{(r)} \mathbf{W}_r^{(\ell)} \mathbf{h}_j^{(\ell)} \right) \tag{2}$$

with $\mathbf{W}_r^{(\ell)}$ and $\alpha_{ij}^{(r)}$ the transformation and attention weights for relation r at layer ℓ and σ a nonlinearity, so that propagation, connectivity, and range relations each learn a distinct interaction rule.

Classical propagation models buy closed-form tractability at the cost of morphological resolution. Free-space path loss follows from the free-space transmission formula [21]; expressed in decibels as a function of distance and frequency, it sets the theoretical attenuation floor,

$$L_{\mathrm{FSPL}} = 20 \log_{10}(d) + 20 \log_{10}(f) + 32.44 \quad [\mathrm{dB}] \tag{3}$$

with d in km and f in MHz; Okumura–Hata adds empirical corrections for antenna height [1], [22] and for suburban and open areas relative to the urban formula [1], and COST 231 [2] carries the same scheme into the 1500–2000 MHz band. Terrain-aware standards go further: ITU-R P.1546 [4] integrates elevation profiles for point-to-area field strength, and ITU-R P.833 [23] models canopy attenuation. Even so, the Okumura–Hata and COST 231 models treat two environments alike whenever they fall into the same propagation category, which limits their accuracy at a given location [7], and a first analysis reports that high-resolution

terrain and land-use data improve ITU-R P.1812 predictions where small obstacles block the signal [24].

Learned estimators raise accuracy over empirical models, but each line of work solves only part of the terrain problem. RadioUNet [11] treats path loss as image translation on a 2D grid and was the public baseline of a challenge on path-loss radio maps from city maps and transmitter locations [25]. Radio-map estimation from sparse measurements [26] keeps that grid in its image-based methods: completion autoencoders take the measurements with a mask of their grid locations and the building map, as surveyed in [7], and RME-GAN [12] generates coverage adversarially from sparse observations placed on a grid.

Graph-based estimators inherit the same grid. A data-driven GNN estimator reconstructs radio environment maps from sparse measurements, with hexagonal tiles linked to their grid neighbors as nodes [13]. RadioGAT [27] reconstructs multi-band radio maps by graph attention over grid cells, linking cells within a distance threshold when their radio depth, derived from a log-distance model and the building-free fraction of the path to each transmitter, is similar, so its propagation model enters through the construction of the graph rather than through the loss. A message-passing GNN trained on measured 4G received power [28] connects pixel nodes through grid edges and, added to them, through edges aligned with the transmitter, the closest counterpart in the cited work to the two forward relations of GNN-RF, though both of its edge sets join pixels to pixels and the transmitter is not a node, and holds out whole sites for validation. Each of these representations therefore still ties its inputs to a regular grid.

Recent raster estimators keep this design. RadioAR [29] takes binary building and transmitter masks of one transmitter per map on a fixed grid, learns from ray-traced RadioMapSeer maps split by whole region, and outputs one path-loss channel without physics in training. Using it here would require rasterizing the graph and replacing a target of path loss and RSSI in decibels, and coverage, so it is not a direct baseline.

Physics-informed networks embed conservation laws as soft penalties for data-efficient learning that respects domain knowledge [30], yet in wireless settings physics in the training objective has rarely reached past a path-loss floor [6], [12], [26], [31]. This work measures instead a graph representation with antenna-to-terrain relations, evaluated in spatial blocks against calibrated empirical models and learned controls, with an output of path loss and RSSI in decibels, and coverage.

A terrain graph is only as informative as the features it carries. Digital elevation models supply slope, aspect, curvature, the Topographic Position Index, and roughness [32], [33], global geomorphometric layers add the Terrain Ruggedness Index [34], and Sentinel-2 indices such as NDVI [35] and NDWI [36] encode vegetation density and water. Outside graph models, geospatial descriptors encode the path-specific obstruction geometry [10], and satellite-image features improve prediction where vegetation and buildings cause fading that rigid empirical models represent poorly [6]. GNN-RF carries eight descriptors computed from the Copernicus DEM and six from Sentinel-2 at every terrain node (Section III-A).

TABLE 1: Graph census and training cost. Counts are read from the 16 quadrant graph files; the terrain grid, and therefore the terrain-to-terrain edge count, is the same in every cell. Antenna nodes are all licensed transmitters of the city: 175 in Lins, 794 in Bauru, 1398 in Sorocaba, 4252 in Campinas. Costs are over the 80 blocked runs on one NVIDIA RTX 5070 Ti (16 GB).

Terrain nodes per cell	12 960 000
Antenna nodes per cell	175–4252
Antenna→terrain edges (and reverse)	4 524 526–10 503 392
Terrain→terrain edges (8-neighbour)	116 596 804
Carrier frequencies	778–3550 MHz
Serialised graph file	28.2–28.3 GB
Sampled neighbours per node (antenna / terrain)	20 / 8
Training batch (seed nodes)	24 576
Epochs (head quadrant / transferred)	35 / 20
Trainable parameters	1 812 515
Peak training VRAM (min–max, median)	7430–7512 MB, 7467
Wall-clock per run (min–max, median)	2181–4032 s, 2292

Operational use also requires calibrated confidence, since average error alone cannot support risk-sensitive decisions. Monte Carlo Dropout [16] approximates Bayesian inference through repeated stochastic passes, and conformal prediction offers distribution-free coverage under exchangeability [37], which in graph models holds for transductive node classification and breaks once a test node enters the receptive field of a calibration node [38]. Uncertainty-aware radio-map estimation nevertheless remains thinly integrated with spatial error analysis, and this work tests how far that gap can be closed by evaluating per-prediction intervals under spatial blocking in every city–quadrant cell it studies.

III. PROPOSED ARCHITECTURE

A. HETEROGENEOUS GRAPH CONSTRUCTION

The RF propagation environment for each city quadrant is represented as a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}, \phi, \psi)$ with two node types and three edge (relation) types, constructed from enriched terrain datasets combined with antenna deployment data. Graph construction is a one-time preprocessing step; the resulting graph is serialized to disk (28.2–28.3 GB per quadrant in PyTorch Geometric format) and reused across all training runs, and Table 1 lists its exact node and edge counts per relation, the sampling fan-outs, and the training budget.

1) Node types and features

Terrain nodes represent DEM grid cells enriched with spectral indices from Sentinel-2 imagery, and antenna nodes the licensed transmitters of each city (Table 1). The 17 raw terrain features are augmented at load time with a normalized

TABLE 2: Terrain node feature vector ($d_t = 18$).

Index	Feature	Source	Category
0	Elevation (z-score)	DEM	Topographic
1	Slope	DEM	Topographic
2	Aspect (cosine)	DEM	Topographic
3	Aspect (sine)	DEM	Topographic
4	Curvature	DEM	Topographic
5	TPI	DEM	Topographic
6	Local elevation range	DEM	Topographic
7	Roughness	DEM	Topographic
8	B02 (Blue)	Sentinel-2	Spectral
9	B03 (Green)	Sentinel-2	Spectral
10	B04 (Red)	Sentinel-2	Spectral
11	B08 (NIR)	Sentinel-2	Spectral
12	NDVI	Sentinel-2	Vegetation
13	NDWI	Sentinel-2	Moisture
14	LiDAR flag	GEDI L2A + ICESat-2 ATL08	Boolean
15	LiDAR elevation (normalized)	GEDI L2A + ICESat-2 ATL08	LiDAR
16	Constant	—	Constant
17	d_{ant} (normalized)	Computed	Geometric

distance-to-nearest-antenna feature, yielding $d_t = 18$ input dimensions per terrain node.

Table 2 lists the 18 terrain node features. Features 0–7 are computed directly from the raw Copernicus GLO-30 DEM at 30 m resolution [39] using standard geomorphometric algorithms (elevation as a z-score; slope from central-difference gradients of the elevation; aspect as its cosine and sine; curvature; TPI as elevation minus the 3×3 mean, the cell-minus-focal-mean index that Wilson et al. [40] compute over a circular neighborhood, here on a square window; local elevation range as the 3×3 range, and roughness as the 3×3 standard deviation of elevation). No learned super-resolution or enhancement is applied to the DEM; the topographic features represent the native Copernicus product quality, including the elevation bias that vegetation height and cover introduce into the Copernicus DEM [41].

Features 8–11 are the four 10 m bands (B02, B03, B04, B08) of Sentinel-2 Level-2A imagery [42], and features 12 and 13 are $\text{NDVI} = (\text{B08} - \text{B04})/(\text{B08} + \text{B04})$ and $\text{NDWI} = (\text{B03} - \text{B08})/(\text{B03} + \text{B08})$. Features 14 and 15 are a LiDAR-availability flag and a normalized LiDAR elevation from GEDI L2A [43] and ICESat-2 [44] ATL08 terrain heights [45] (NASA Earthdata, 2022–2023), non-zero on a small share of nodes; feature 16 is a constant column, so 17 of the 18 inputs are informative; and feature 17 is the z-score-normalized Haversine distance to the nearest antenna, computed from geographic coordinates.

The distance feature is computed from the enriched terrain graph:

$$d_{\text{ant},i} = \min_{j \in \mathcal{V}_a} d_{\text{Haversine}}(\mathbf{p}_i, \mathbf{p}_j), \qquad \bar{d}_{\text{ant},i} = \frac{d_{\text{ant},i} - \mu_d}{\max(\sigma_d, 1.0)} \tag{4}$$

where $\mathcal{V}_a$ is the antenna node set and μ_d, σ_d are the per-graph mean and standard deviation.

2) Edge types

Three relation types connect the graph. Antenna-to-terrain edges encode propagation paths; terrain-to-terrain edges capture 8-connected spatial adjacency from the DEM grid (each interior node connects to its 8 immediate neighbors); and the reverse relation exists in the graph but carries no message in training or inference. Antenna-to-terrain edges carry the link distance and the carrier frequency of the transmitter, each scaled by a fixed constant. Terrain-to-terrain edges carry a 2-dimensional attribute vector $\mathbf{e}_{ij} = [d_{ij},\ \Delta z_{ij}]$, where d_{ij} is the grid distance between nodes i and j (30 or 42.4 m, and 0 on the self-loop of each node) and Δz_{ij} is the difference of their standardized elevations, both used without further normalization.

3) Prediction targets

Each terrain node is associated with five RF targets: total path loss L_{total} (dB), vegetation attenuation L_{veg} (dB), terrain diffraction loss L_{ter} (dB), received signal strength indicator RSSI (dBm), and coverage probability $p_{\text{cov}} \in [0, 1]$.

B. GNN-RF ENCODER–DECODER ARCHITECTURE

1) Terrain encoder

The terrain encoder (Fig. 1) maps raw features $\mathbf{x}_i^t \in \mathbb{R}^{18}$ to hidden representations:

$$\mathbf{h}_i^{t,(0)} = f_{\text{BN}}\big(\mathbf{W}_2^t \cdot \sigma(\text{BN}(\mathbf{W}_1^t \mathbf{x}_i^t + \mathbf{b}_1^t)) + \mathbf{b}_2^t\big) \tag{5}$$

where $\mathbf{W}_1^t \in \mathbb{R}^{d_h \times d_t}$, $\mathbf{W}_2^t \in \mathbb{R}^{d_h \times d_h}$, BN denotes batch normalization, $\sigma = \text{LeakyReLU}(0.2)$, and dropout with rate 0.1 is applied after the first activation [46].

2) Antenna encoder

The antenna encoder maps $\mathbf{x}_j^a \in \mathbb{R}^6$ through a three-layer MLP:

$$\mathbf{h}_j^{a,(0)} = f_{\text{BN}}\big(\mathbf{W}_3^a \cdot \sigma(\mathbf{W}_2^a \cdot \sigma(\text{BN}(\mathbf{W}_1^a \mathbf{x}_j^a + \mathbf{b}_1^a)) + \mathbf{b}_2^a) + \mathbf{b}_3^a\big) \tag{6}$$

with intermediate dimension 128 and output $\mathbf{h}_j^{a,(0)} \in \mathbb{R}^{d_h}$.

3) Heterogeneous GATv2 layers

The encoder stacks L=4 heterogeneous convolutional layers, each a `HeteroConv` module that routes a distinct convolution to every relation. The two forward relations, `propagates_to` and `connects_to`, use GATv2Conv [19] with K=4 heads and per-head dimension d_h/K=64, consuming the 2D edge attributes so that attention can weight neighbors by geometry. The reverse relation `in_range_of` (terrain→antenna) uses the cheaper SAGEConv with mean aggregation [47], since antenna states are never decoded into predictions and the relation carries as many edges as the forward antenna-to-terrain relation (up to 10 503 392 per quadrant, Table 1). Under the one-hop neighbor sampler used for training and inference, however, the reverse relation has no edges in any batch.

Each layer applies LeakyReLU activation, dropout with rate 0.1, residual connections (when dimensions match) [48], and per-type LayerNorm:

$$\mathbf{h}_i^{(\ell)} = \text{LN}\Big(\text{Drop}\big(\sigma\big(\text{HetConv}^{(\ell)}(\mathbf{h}^{(\ell-1)})\big)\big) + \mathbf{h}_i^{(\ell-1)}\Big) \tag{7}$$

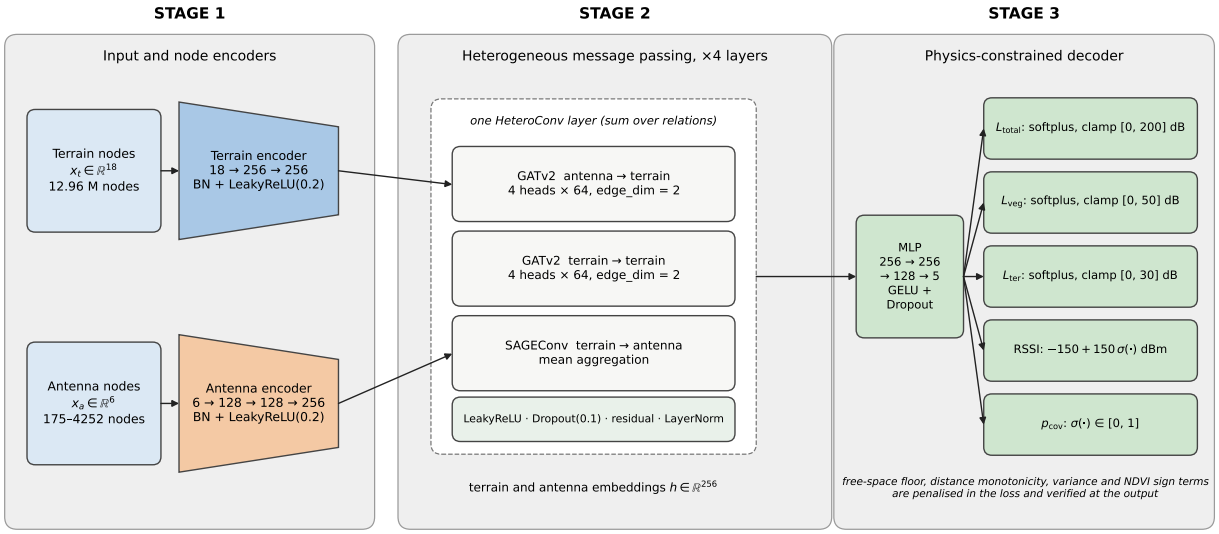


FIGURE 1: GNN-RF encoder–decoder architecture: type-specific terrain and antenna encoders, four heterogeneous GATv2 layers with residual connections, and a physics-constrained decoder with five RF outputs (hyperparameters in Table 3).

4) Physics-constrained decoder

The decoder maps terrain embeddings $\mathbf{h}_i^{(L)} \in \mathbb{R}^{d_h}$ to five predictions through an MLP with GELU activations:

$$\hat{\mathbf{y}}_i = \mathbf{W}_3^d \cdot \mathrm{GELU}(\mathbf{W}_2^d \cdot \mathrm{GELU}(\mathbf{W}_1^d \mathbf{h}_i^{(L)})) \tag{8}$$

with dimensions $d_h \to d_h \to d_h/2 \to 5$ and dropout with rate 0.1 after each GELU.

Physics constraints are enforced via output activation functions:

$$\hat{L}_{\text{total}} = \min\big(\mathrm{softplus}(\hat{y}_0),\ 200\big) \tag{9}$$
$$\hat{L}_{\text{veg}} = \min\big(\mathrm{softplus}(\hat{y}_1),\ 50\big) \tag{10}$$
$$\hat{L}_{\text{ter}} = \min\big(\mathrm{softplus}(\hat{y}_2),\ 30\big) \tag{11}$$
$$\widehat{\text{RSSI}} = -150 + 150 \cdot \sigma(\hat{y}_3) \quad \in [-150, 0]\ \text{dBm} \tag{12}$$
$$\hat{p}_{\text{cov}} = \sigma(\hat{y}_4) \quad \in [0, 1] \tag{13}$$

where $\mathrm{softplus}(x) = \ln(1 + e^x)$ ensures non-negative path loss and σ denotes the sigmoid function.

C. PHYSICS-INFORMED LOSS FUNCTION

The composite loss function combines data-driven regression with four physics-informed regularization terms. All five output channels and all four regularizers are active from the first epoch, with no training phases.

1) Prediction loss

The primary loss uses Huber regression [49] with δ=5.0 and channel weights:

$$\mathcal{L}_{\text{pred}} = \sum_{c=0}^{4} w_c \cdot \mathcal{L}_{\text{Huber}}(\hat{y}_c, y_c; \delta{=}5.0). \tag{14}$$

TABLE 3: GNN-RF architectural hyperparameters. Antenna input: carrier frequency, transmit power, height, gain, latitude, longitude.

Parameter	Value
Hidden dimension (d_h)	256
Number of GNN layers (L)	4
Attention heads (K)	4
Edge feature dimension	2
Output dimension	5
Dropout rate	0.1
Terrain input dimension (d_t)	18 (17 informative)
Antenna input dimension (d_a)	6
Neighbor fan-out (antenna / terrain)	20 / 8
Training batch (seed nodes)	24 576
Total parameters	1 812 515

In the training script used for every reported run the channel weights stay uniform ($w_c = 0.2$) and all five channels are active in every epoch.

2) FSPL lower-bound constraint

Total path loss must not fall below the theoretical free-space minimum (Eq. (3)):

$$\mathcal{L}_{\text{FSPL}} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{ReLU}\big(L_{\text{FSPL},i} - \hat{L}_{\text{total},i}\big) \tag{15}$$

This term penalizes violations in the loss; unlike the bounds of Eqs. (9)–(13), it is not imposed by the output activation, and admissibility is verified on the predictions in Section V-D.

3) Distance–attenuation monotonicity

RF attenuation generally increases with distance from the transmitter. We encourage this prior as a soft regularizer by

sampling M=512 random node pairs (i, j) where $d_i < d_j$:

$$\mathcal{L}_{\text{dist}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \text{ReLU}\big(\widehat{\text{RSSI}}_j - \widehat{\text{RSSI}}_i + 1\big) \tag{16}$$

where $\mathcal{P} = \{(i, j) : d_{\text{ant},i} < d_{\text{ant},j}\}$. The 1 dB margin imposes a minimum decrease of RSSI from the node closer to the nearest antenna to the farther one, and the term is zero only when that decrease is reached. Because the reference field takes the maximum received level over all transmitters, the relation to the nearest antenna holds only on average, and the term is a regularizer rather than a property of the field.

4) Variance preservation

Predicted RSSI should preserve the spatial variance of the target distribution:

$$\mathcal{L}_{\text{var}} = \big[\text{ReLU}(\sigma_{y_3} - \sigma_{\hat{y}_3})\big]^2 \tag{17}$$

5) Vegetation–NDVI sign consistency

Predicted vegetation attenuation should correlate positively with observed vegetation density:

$$\mathcal{L}_{\text{NDVI}} = \text{ReLU}\big(0.15 - \rho(\hat{L}_{\text{veg}}, \text{NDVI})\big) \tag{18}$$

where ρ denotes Pearson correlation and 0.15 is a minimum-correlation threshold. The term penalizes only a wrong-signed or negligible relation between predicted vegetation attenuation and NDVI, a sign condition motivated by, but not equivalent to, the vegetation attenuation model of ITU-R P.833 [23].

6) Total loss

The complete loss is:

$$\begin{aligned}\mathcal{L} = \sum_{c=0}^{4} w_c \mathcal{L}_{\text{Huber}}^{(c)} + \lambda_F \mathcal{L}_{\text{FSPL}} + \lambda_V \mathcal{L}_{\text{veg}} \\ + \lambda_d \mathcal{L}_{\text{dist}} + \lambda_v \mathcal{L}_{\text{var}} + \lambda_n \mathcal{L}_{\text{NDVI}}\end{aligned} \tag{19}$$

where $(\lambda_F, \lambda_V, \lambda_d, \lambda_v, \lambda_n) = (0.05, 0.1, 0.05, 0.02, 0.03)$. The vegetation term $\mathcal{L}_{\text{veg}}$ is a Huber penalty between $\hat{L}_{\text{veg}}$ and a simplified ITU-R P.833 attenuation computed from canopy height; no canopy-height layer is available in this study, so this term is inactive in every reported run, and the vegetation and terrain-diffraction channels are supervised only through their Huber terms in Eq. (14) against the corresponding components of the reference generator, documented in the Supplementary Material.

D. SEQUENTIAL TRANSFER LEARNING PROTOCOL

Training follows a sequential transfer protocol across four geographic quadrants per city (Algorithm 1). Q1 trains from random initialization for 35 epochs at learning rate 10^{-3}; subsequent quadrants fine-tune [50] from the preceding best checkpoint for 20 epochs at 2×10^{-4}, resetting the optimizer and scheduler state. The cosine schedule decays to $\eta_{\min}$=10^{-6}, and float16 mixed precision uses a GradScaler. Section V-B compares this schedule with training every quadrant from scratch.

Algorithm 1 Sequential Transfer Learning Protocol

Require: City c, seed s, quadrant graphs $\{G_{Q_1}, \ldots, G_{Q_4}\}$
Initialize model $\theta \sim \mathcal{N}(0, \text{fan-in}^{-1})$
for $q \in \{Q_1, Q_2, Q_3, Q_4\}$ **do**
  if $q = Q_1$ **then**
    $\eta \leftarrow 10^{-3}$; $E \leftarrow 35$
  else
    $\theta \leftarrow \theta^*_{q-1}$ {load best checkpoint}
    $\eta \leftarrow 2 \times 10^{-4}$; $E \leftarrow 20$
  end if
  optimizer $\leftarrow$ AdamW$(\theta, \eta, \lambda_{\text{wd}}$=$10^{-5})$
  scheduler $\leftarrow$ CosineWarmRestarts$(T_0$=$10, T_{\text{mult}}$=$2)$
  for epoch $= 1$ **to** E **do**
    Train with NeighborLoader (batch size $= 24\,576$)
    Compute $\mathcal{L}$ (Eq. (19)) under AMP (float16)
    $\nabla\theta \leftarrow \text{clip}(\nabla\theta, \|\cdot\|_{\max}$=$0.5)$
    Update θ; step scheduler
  end for
  $\theta^*_q \leftarrow \arg\min_{\text{epoch}} \text{MAE}_{\text{val}}$
end for
return $\{\theta^*_{Q_1}, \ldots, \theta^*_{Q_4}\}$

IV. EXPERIMENTAL METHODOLOGY

A. STUDY AREA AND DATASETS

Experiments cover four cities in São Paulo state, Brazil: Bauru, Campinas, Lins, and Sorocaba, representing diverse urban morphologies from compact urban cores (Campinas) to mixed rural–urban landscapes (Lins). Each city is partitioned into four geographic quadrants (Q1–Q4), yielding 16 city–quadrant evaluation units. Transmitter parameters, with carrier frequencies of 778–3550 MHz and transmit powers of 33.0–53.0 dBm, come from the ANATEL licensing extract documented in the Supplementary Material. The generator reads only frequency, transmit power, antenna height, gain, and site coordinates from this extract and applies gain as a scalar in every direction. Azimuth, mechanical and electrical downtilt, half-power beamwidth, front-to-back ratio, and polarization are present in the same records but never consumed. Every physical tower therefore collapses into a stack of co-located omnidirectional sources that differ only in frequency, power, gain, and height, averaging 13.5 transmitters per tower in Lins and up to 49 in Campinas.

Every accuracy, control, and uncertainty result in this work is obtained under a spatially blocked partition [14]. Each quadrant graph is tiled into 10 km blocks separated by a 2 km exclusion buffer and assigned 70/15/15 by block to training, validation, and test with a fixed partition seed (42). Each partition is a graph of its own, holding only its terrain nodes and copies of all antenna nodes, so that no training terrain node is reachable from validation or test terrain through an antenna. The neighbor sampler draws one hop from terrain seed nodes, so no batch contains terrain-to-antenna edges, and each antenna state depends only on the antenna's own attributes.

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Each cell holds 132 blocks (92 training, 20 validation, 20 test); the buffer, which trims validation and test only, discards 14.7–14.8% of the tile nodes and leaves 817 601 to 833 593 test nodes per cell (Fig. 2). The checkpoint of each run is selected on validation alone, and the test partition is scored once. All 16 cells are trained with five seeds (42–46), for 80 runs. Results quoted from the preliminary experiment, which used a random 90/10 node partition, are identified as such where they appear.

The reference fields are produced by a deterministic propagation generator written for this study and documented equation by equation in the Supplementary Material: an Okumura–Hata base loss [1] up to 1500 MHz and COST 231 [2] above, a vegetation term driven by the node's NDVI, a terrain term driven by the node's slope, a two-term link budget, a best-server rule over all transmitters, a -110 dBm sentinel for unreached nodes, and coverage defined as RSSI above -85 dBm. The generator is point-local and traces no terrain profile between antenna and node. Recomputing the released total path loss from the features stored in the same files reproduces it with a mean absolute error of at most 3.02×10^{-4} dB per cell over 150 000 valid-target nodes from five cells in the four cities, and every error reported in this paper measures fidelity to this generator.

Environment class is assigned once per transmitter from carrier frequency alone, not from a land-cover layer: urban above 1800 MHz, suburban between 800 and 1800 MHz, rural at or below 800 MHz. The best-server rule then searches every transmitter within a 30 km inclusion radius and assigns each node to whichever one maximizes the received level, with no handover hysteresis or load-balancing offset, so the label describes the transmitter's band, not the ground beneath the node. Under these rules, of the 13 186 118 blocked test nodes of the 16 cells, 2 247 437 (17.04%) are valid-target nodes, whose target path loss is valid rather than a sentinel value, and 182 202 (1.38% of test nodes, 8.11% of valid-target nodes) have a target RSSI ≥ -85 dBm, the coverage threshold.

The two auxiliary channels, vegetation attenuation and terrain diffraction, are computed in a later step than total path loss and RSSI, and the terrain term was corrected after generation. An early version of the generator filled each node's slope by an independent draw from a uniform distribution between 0.02 and 0.15 rather than reading the DEM; a corrector later replaced that draw with the node's actual DEM slope and recomputed total loss and RSSI by exact difference. Every field used in this paper's spatially blocked evaluation carries the corrected slope, while results quoted from the preliminary experiment predate the correction and use the uncorrected fields.

B. CLASSICAL BASELINES

Three analytical models serve as lower-bound references: FSPL (Eq. (3)), Okumura–Hata Rural, that is, the open-area case of [1], and COST 231 Suburban [2]. Each is computed from the antenna parameters, distances, and frequencies of the graph data, calibrated on the training blocks of each cell only, and scored in absolute decibels on the same test nodes as GNN-RF, with the -110 dBm floor applied. Okumura–Hata and COST 231 are kept as references even where the carrier lies outside their nominal bands [1], [2].

Each of the three models is evaluated under three calibration variants: (i) a single 900 MHz reference frequency with the offset set to the training median, (ii) the same reference with predicted RSSI clamped to the -110 dBm floor and the offset fitted on the clamped error, and (iii) clamped with the per-link carrier frequency replacing the single 900 MHz reference, yielding nine calibrated candidates per cell. At a fixed frequency the Hata Rural and COST 231 Suburban formulas differ by a constant that the calibrated offset absorbs, so their calibrated predictions coincide under variants (i) and (ii) and diverge only under variant (iii).

C. LEARNED BASELINE: AN IN-HOUSE RADIOUNET-STYLE ADAPTATION

The learned baseline, hereafter the node-wise adaptation, is an in-house encoder–decoder that carries the U-Net-style skip connections of RadioUNet [11] into a fully connected network applied to each terrain node, with widths of 256, 128, and 64 and without convolution, raster, or terrain-to-terrain step. It shares the physics-constrained decoder and the loss of GNN-RF. Each terrain node supplies 44 inputs $(18+5\times5+1)$: the 18 terrain inputs of GNN-RF; for each of up to five antenna-to-terrain links, ordered by increasing distance, the link distance and carrier frequency carried by the antenna-to-terrain edges of the graph, the effective isotropic radiated power (EIRP) of the transmitter, the antenna height, and a presence flag; and $\log(1+n)$ for n links. The baseline consumes the antenna-to-terrain relations of the graph but receives neither the antenna coordinates nor any terrain-to-terrain relation.

With 369 827 trainable parameters, about one fifth of the 1 812 515 of GNN-RF, the baseline is trained in 80 runs (four cities, four quadrants, seeds 42–46) with its own transfer chain. It shares the blocked split, the test nodes, the checkpoint rule, and the epoch budget of the GNN-RF reference runs, and it takes their optimization settings without a hyperparameter search. Each baseline run is paired with the reference run of the same city, quadrant, and seed.

D. ABLATION DESIGN

The ablation asks whether the relations of the graph, rather than parameter capacity, carry the accuracy of GNN-RF, by comparing it with the graph-free control, which receives no graph relation, and with the node-wise adaptation, which receives the antenna-to-terrain relations without message passing. The graph-free control replaces the architecture with a multilayer perceptron matched to GNN-RF in nominal parameters, holding exactly 1 812 515 trainable parameters and no antenna-to-terrain or terrain-to-terrain link, trained with the same loss, partition, checkpoint rule, and epoch budget on all four cities with five seeds each (16 cells, 80 paired

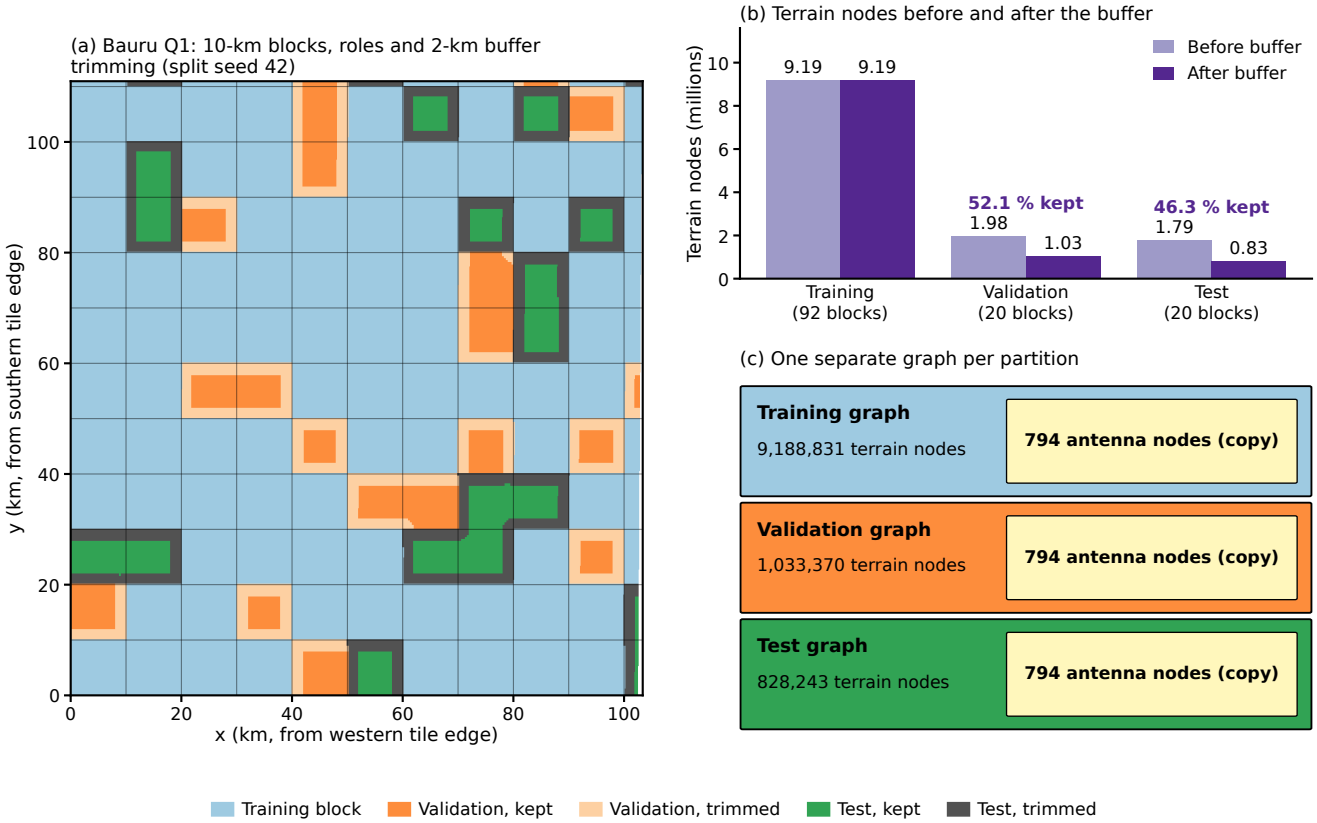


FIGURE 2: Spatially blocked partition, Bauru Q1. (a) 132 blocks of 10 km (92 training, 20 validation, 20 test; split seed 42) trimmed by a 2 km buffer. (b) Terrain nodes before and after the buffer. (c) Each partition is its own graph with a copy of all 794 antenna nodes, so no training terrain node is reachable from validation or test.

runs). All of its parameters are active, whereas GNN-RF has 1 484 037 effective trainable parameters, a mismatch that favors the control and makes the comparison conservative where GNN-RF has the lower error.

A second arm keeps the trained seed-42 models and damages the graph at test time on the eight cells of Lins and Bauru, with three shuffling seeds per arm: antenna-to-terrain edges are permuted preserving node degree, terrain-to-terrain edges are permuted the same way, both are permuted together, and message passing is removed by evaluating on an empty graph.

Two retrained arms test the objective: one (C1) sets the weights of all four penalty terms to zero, and the other (C2) sets only the monotonicity weight to zero, keeping the FSPL-floor, variance, and NDVI weights of the reference runs. Each arm has 80 runs with its own transfer chain, copies every other training setting from the corresponding reference run, and uses the same blocked split, test nodes, and evaluation, with its differences from the reference paired as for the learned baseline. The FSPL-floor penalty is essentially inactive in training, its weighted value never exceeding 6.06×10^{-4} of the weighted RSSI data term in any epoch, so it has no retraining arm of its own. The variance and NDVI terms have no separate arms; C1 removes them together with the other two terms and is inconclusive relative to the prespecified margin in both populations.

E. EVALUATION METRICS AND STATISTICAL TESTS

Primary metrics:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} |\hat{y}_i - y_i| \tag{20}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (\hat{y}_i - y_i)^2} \tag{21}$$

$$\mathrm{P90} = Q_{0.9}(|\hat{y}_i - y_i|) \tag{22}$$

The unit of analysis for claims of geographic generalization is the city (n=4), with the cell (n=16, seeds as replicates) and the seed chain as secondary readings; terrain nodes are never the unit of a test, and the intraclass correlation of cell by seed is 0.73, so runs are not treated as independent either. Every results table names its node population (all test nodes or valid-target nodes) and reports the mean of cells unless stated as pooled; each run enters through its checkpoint of lowest validation RSSI MAE, and inference

uses the neighbor-sampling evaluator with 20 antenna and 8 terrain neighbors per edge type.

Confidence intervals are computed over cells. Paired contrasts report the mean difference with a 95% t interval and the paired t and Wilcoxon [51], [52] p-values and, where absence of an effect is the question, two one-sided tests (TOST) at a $0.1\,\mathrm{dB}$ margin fixed before the runs; comparisons among cities use Wilcoxon tests with Holm correction. Each seed enters through the mean of its cell, and where the city is the unit the interval is computed over the city means, without resampling at any level.

For the retrained arms and the learned baseline, each difference is the mean of 80 run-level differences paired by city, quadrant, and seed, with a 95% interval from a hierarchical bootstrap (B=20 000) that resamples cities and, within each city, seed chains with their four quadrants together. Differences are given as the arm or model minus GNN-RF, negative where the arm or model has the lower error. Such a difference is inconclusive relative to the pre-specified margin when its interval crosses that margin, fixed before the retrained and baseline runs at $\pm 0.045\,\mathrm{dB}$ for the MAE over all test nodes, $\pm 0.35\,\mathrm{dB}$ for the MAE over valid-target nodes, and $\pm 0.1\,\mathrm{dB}$ for the valid-target bias of the retrained arms. Each MAE margin is the minimum detectable effect at 80% power of a paired two-sided t test (α=0.05, 16 city–quadrant cells) implied by the seed-to-seed variability of the 80 reference GNN-RF runs.

For uncertainty, prediction intervals are built at nominal levels $q \in \{50, 80, 90, 95, 99\%\}$ and scored by:

$$\mathrm{PICP}_q = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\big[y_i \in [\hat{y}_i^{\mathrm{lo}}(q), \hat{y}_i^{\mathrm{hi}}(q)]\big] \tag{23}$$

$$\mathrm{Width}_q = \frac{1}{N}\sum_{i=1}^{N} \big(\hat{y}_i^{\mathrm{hi}}(q) - \hat{y}_i^{\mathrm{lo}}(q)\big) \tag{24}$$

together with interval score, negative log-likelihood, and CRPS. The calibration region for split-conformal and deterministic-residual intervals is the validation blocks of the same cell, spatially separated from its test blocks, and a method is called calibrated only if $|\mathrm{PICP}_q - q| \le 0.05$ at $q \in \{80, 90, 95\%\}$ in at least 12 of the 16 cells, a criterion fixed before the evaluation.

F. COMPUTATIONAL INFRASTRUCTURE

All experiments ran on a workstation with a single NVIDIA RTX 5070 Ti GPU (16 GB VRAM), using PyTorch 2.10 with CUDA 12.8 and PyTorch Geometric. Across the 80 blocked runs, peak training VRAM is 7430–7512 MB (median 7467 MB) and wall-clock time per run is 2181–4032 s (median 2292 s).

V. RESULTS AND ANALYSIS

Table 4 lists the MAE values and paired MAE differences reported in this section, with the protocol and seeds behind each.

TABLE 4: Results roadmap: MAE values and paired MAE differences of Section V, all from the spatially blocked protocol with five seeds per cell except the empty-graph ablation, MC-mean, and cross-city rows (seed 42). Rows reading "X minus GNN-RF" give the paired difference in MAE, negative where X has the lower error; intervals and margins of the node-wise adaptation and of the retrained arms (C1, C2) follow Section IV-E.

Context	Scope	MAE (dB)	95% CI (dB)
Blocked test, all nodes	16 cells	0.395	
Blocked test, valid-target nodes	16 cells	2.012	
Node-wise adaptation minus GNN-RF, all nodes	16 cells	−0.0571	[−0.1183, +0.0025]
Node-wise adaptation minus GNN-RF, valid-target nodes	16 cells	−0.7794	[−1.8484, −0.1163]
Empty-graph ablation minus GNN-RF	8 cells	+1.72	
Graph-free control minus GNN-RF	16 cells	+0.247	
No-penalty retraining (C1) minus GNN-RF, all nodes	16 cells	−0.0593	[−0.1249, +0.0156]
No-penalty retraining (C1) minus GNN-RF, valid-target nodes	16 cells	+0.1497	[−0.3082, +0.8106]
No-monotonicity retraining (C2) minus GNN-RF, all nodes	16 cells	−0.0865	[−0.1323, −0.0391]
No-monotonicity retraining (C2) minus GNN-RF, valid-target nodes	16 cells	−0.0768	[−0.3512, +0.3093]
MC-mean, valid-target nodes	16 cells	2.35	
Cross-city, full/zeroed topography	3 cities	0.40–3.93	

A. EVALUATION PROTOCOL AND PARTITION LEAKAGE

A nearest-neighbor predictor exposes partition leakage on this reference field in all 16 cells. Restricting the copied training neighbor to a node with a valid target and sampling up to 20 000 valid-target test nodes per cell (seed 42), Table 5 reports its path-loss MAE under both partitions in every city–quadrant cell, with a median of 0.41 dB under the random node split and 4.14 dB under the spatially blocked split. The blocked-split MAE exceeds the random-split MAE in all 16 cells, with a median ratio of 10.5 (range 3.1–40.3) and a median gap of 3.80 dB (range 1.52–6.45 dB); the pre-specified presence criterion holds in all 16 cells and all four cities. A trivial copy of the nearest labeled neighbor is therefore adequate under random partitioning and fails under spatial blocking [53], so the partition rather than the predictor drives the gap.

This contrast uses a single partition seed and a sample of up to 20 000 test nodes per cell, and its nearest-neighbor tie-break rule, inherited from the spatial index, was tested against an alternative rule in only 3 of the 16 cells (Bauru Q3, Sorocaba Q2, Lins Q1) without a change in conclusion. The blocked test set holds 6.3–6.4% of the tile after the buffer, and the two splits differ in the valid-target population of the test set.

In the preliminary experiment, label-shuffled controls

TABLE 5: Nearest-neighbour (1-NN) path-loss predictor under a random and a spatially blocked node partition, 16 city–quadrant cells (seed 42). The prediction for a test node is the path loss of the nearest training node with a valid path loss; MAE is computed on up to 20 000 sampled test nodes with a valid path loss per cell. The random partition has the same training and test sizes as the blocked partition. Ratio is blocked over random MAE in each cell; summary rows give the median and range of each column. Cells share a seed and adjacent tiles, so medians and ranges are shown instead of confidence intervals.

		1-NN MAE (dB)		
City	Q	Random	Blocked	Ratio
Bauru	Q1	0.39	6.65	17.2
Bauru	Q2	0.26	4.23	16.2
Bauru	Q3	0.49	2.01	4.1
Bauru	Q4	0.39	4.12	10.5
Campinas	Q1	0.35	4.76	13.6
Campinas	Q2	0.87	2.89	3.3
Campinas	Q3	0.55	4.17	7.6
Campinas	Q4	0.99	4.92	4.9
Lins	Q1	0.19	3.08	16.6
Lins	Q2	0.16	6.61	40.3
Lins	Q3	0.24	4.11	17.4
Lins	Q4	0.21	2.18	10.4
Sorocaba	Q1	0.44	5.10	11.7
Sorocaba	Q2	0.87	2.73	3.1
Sorocaba	Q3	0.77	7.03	9.2
Sorocaba	Q4	1.08	4.05	3.7
Bauru	median	0.39	4.17	13.4
	range	0.26–0.49	2.01–6.65	4.1–17.2
Campinas	median	0.71	4.46	6.3
	range	0.35–0.99	2.89–4.92	3.3–13.6
Lins	median	0.20	3.60	17.0
	range	0.16–0.24	2.18–6.61	10.4–40.3
Sorocaba	median	0.82	4.57	6.5
	range	0.44–1.08	2.73–7.03	3.1–11.7
All cells	median	0.41	4.14	10.5
	range	0.16–1.08	2.01–7.03	3.1–40.3

raised RSSI MAE by factors of 3.29 to 10.35 in all 16 cells, whereas the geographic-split control, measured on a RadioUNet adaptation of that experiment, passed its own threshold in only 8 of 16 cells, failing in Q1 and Q3 of every city. That failure motivates the spatially blocked protocol of Section IV adopted throughout this paper.

B. MODEL COMPARISON AND BASELINES

Under the spatially blocked protocol, GNN-RF is scored against empirical models calibrated on the training blocks (Tables 4 and 6). Over all test nodes, mean RSSI MAE is $0.395\,\mathrm{dB}$ (95% CI [0.309, 0.480], n=16 cells; [0.304, 0.485] with the city as unit), ranging from $0.215\,\mathrm{dB}$ in Lins Q4 to $0.812\,\mathrm{dB}$ in Campinas Q3, whereas under calibration variant (iii) (per-link carrier, -110 dBm floor), calibrated FSPL reaches 0.74–3.09 dB and calibrated Hata Rural and COST 231 Suburban 0.35–2.72 dB on the same nodes. Over all test nodes, against the best calibrated empirical model of each cell, GNN-RF is more accurate in 15 of 16 cells and trails by $0.009\,\mathrm{dB}$ in Campinas Q2. Over the 2 247 437 valid-target nodes, GNN-RF reaches $2.012\,\mathrm{dB}$ as the mean of cells and $1.665\,\mathrm{dB}$ pooled by node (cells weighted by their valid-target nodes), against $4.362\,\mathrm{dB}$ and $4.935\,\mathrm{dB}$, respectively, for the best calibrated empirical model of each cell on those nodes.

A paired test confirms this margin: against the best of nine calibrated analytical candidates in each cell, GNN-RF reduces mean RSSI MAE by $0.746\,\mathrm{dB}$ (95% CI [0.411, 1.080], p=2.59×10^{-4} paired t, p=6.10×10^{-5} Wilcoxon, Cohen's d_z=1.19). With the city as the unit (n=4), the margin is $0.746\,\mathrm{dB}$ (95% t interval [0.331, 1.161], p=0.0106 paired t; Wilcoxon p=0.125, the minimum attainable with four cities), and GNN-RF is more accurate in all four cities. The 16 cells are not strictly independent, since quadrants of the same city are adjacent tiles, and selecting the best analytical candidate per cell is an optimistic envelope for the baseline, which is conservative for this claim.

On valid-target nodes, where the generator assigns a valid path loss, the margin widens: GNN-RF reaches 0.82–9.65 dB RSSI MAE over all 16 cells, Lins Q1 included, and 2.801 dB path-loss MAE as the mean of cells, against 1.12–12.03 dB RSSI MAE for the best empirical model. It trails in four cells: Lins Q1, where only 3% of test nodes are valid-target (9.65 against 3.69 dB), Lins Q2 (2.64 against 1.65 dB), Campinas Q2 (1.71 against 1.30 dB), and Sorocaba Q2 (1.84 against 1.12 dB). A constant -110 dBm predictor, scored on all test nodes, never comes within 0.5 dB of GNN-RF in any of the 16 cells, so the all-node figure does not reduce to predicting the sentinel.

Table 7 tests the sequential schedule of Section III-D against training each non-head quadrant from scratch on the same blocked split. Over 24 pairs (four cities, seeds 42 and 43, Q2–Q4), the chained model differs from the from-scratch model by $-0.014\,\mathrm{dB}$ (95% CI [-0.072, $+0.045$], p=0.60, eight city–seed clusters), and at this unit two one-sided tests exclude a difference larger than $\pm0.1\,\mathrm{dB}$ (p=0.0051). With the city as the unit of analysis (n=4, seeds and quadrants nested within cities), the chained-minus-scratch difference in RSSI MAE is $-0.014\,\mathrm{dB}$ (95% CI [-0.107, $+0.079$], t with 3 degrees of freedom). This interval crosses the pre-specified $\pm0.1\,\mathrm{dB}$ margin, so no systematic difference is detectable at the city unit (TOST p=0.030; minimum detectable difference at 80% power, 0.12 dB).

Two features of the chain qualify this comparison. Because the from-scratch runs use 35 epochs at 10^{-3} and the chained runs 20 epochs at 2×10^{-4}, the comparison detects no systematic advantage of chaining but does not give a clean estimate of its effect; a reversed chain Q4→Q1 in Lins, on one seed, gives 0.392 against 0.399 dB. The chain also brings the training blocks of one quadrant close to the test nodes of the next: in each Q2 cell, 10.97 to 11.07% of the valid-target test nodes lie within 2 km of the Q1 training blocks, and excluding them changes the valid-target RSSI MAE of

TABLE 6: Test error on the spatially blocked split, all 16 city–quadrant cells, three populations of test nodes: all nodes (sentinel included), nodes with a valid path-loss target ("valid-target"), and the valid-target subset for path loss. GNN-RF entries are mean ± s.d. over training seeds; MLP is the graph-free control of Section IV-D (all nodes; mean over training seeds). The node-wise adaptation is the in-house adaptation of RadioUNet described in Section IV-C (all nodes and valid-target nodes; mean ± s.d. over training seeds). "Best empirical" is the lowest-error calibrated empirical model on the same nodes, chosen separately for the all-node and valid-target populations; "Constant" is the −110 dBm sentinel predictor. Every entry uses the checkpoint of lowest validation RSSI MAE and the test nodes of its population; each cell is the mean of five seeds, the last row is the unweighted mean of the 16 cells, and the GNN-RF evaluator is stochastic, sampling neighbors ($k_{\mathrm{antenna}} = 20$, $k_{\mathrm{terrain}} = 8$).

			GNN-RF (dB)			Node-wise adaptation (dB)		MLP	Best empirical (dB)		Constant
City	Q	Valid-target frac.	RSSI, all	RSSI, valid-target	PL, valid-target	RSSI, all	RSSI, valid-target	RSSI, all	RSSI, all	RSSI, valid-target	RSSI, all
Bauru	Q1	0.328	0.632 ± 0.093	1.67 ± 0.34	2.21 ± 0.54	0.565 ± 0.126	1.41 ± 0.35	0.927	1.932	3.99	5.236
Bauru	Q2	0.150	0.369 ± 0.073	1.62 ± 0.34	2.82 ± 0.65	0.442 ± 0.086	2.13 ± 0.48	0.373	0.982	4.33	1.694
Bauru	Q3	0.159	0.391 ± 0.048	1.54 ± 0.30	2.14 ± 0.66	0.230 ± 0.012	0.57 ± 0.09	0.434	1.019	3.82	2.613
Bauru	Q4	0.135	0.236 ± 0.071	1.09 ± 0.43	2.20 ± 1.10	0.173 ± 0.053	0.55 ± 0.22	0.205	0.806	3.61	2.077
Campinas	Q1	0.241	0.495 ± 0.100	1.63 ± 0.45	3.76 ± 1.66	0.724 ± 0.155	2.62 ± 0.74	1.824	1.994	7.38	2.417
Campinas	Q2	0.151	0.350 ± 0.205	1.71 ± 1.16	2.90 ± 3.00	0.254 ± 0.048	1.00 ± 0.49	0.315	0.341	1.30	2.636
Campinas	Q3	0.297	0.812 ± 0.170	2.37 ± 0.54	3.55 ± 1.54	0.658 ± 0.071	1.74 ± 0.18	1.229	2.035	6.86	3.882
Campinas	Q4	0.135	0.222 ± 0.078	1.12 ± 0.56	2.16 ± 1.51	0.161 ± 0.025	0.45 ± 0.11	0.568	0.787	2.98	2.087
Lins	Q1	0.030	0.447 ± 0.072	9.65 ± 2.22	8.65 ± 2.35	0.248 ± 0.056	3.42 ± 1.00	0.366	1.200	3.69	1.130
Lins	Q2	0.148	0.474 ± 0.065	2.64 ± 0.49	2.40 ± 0.59	0.202 ± 0.017	0.54 ± 0.13	0.172	0.484	1.65	4.334
Lins	Q3	0.100	0.254 ± 0.049	0.95 ± 0.44	1.63 ± 0.64	0.231 ± 0.034	0.55 ± 0.22	0.227	0.436	3.02	1.830
Lins	Q4	0.125	0.215 ± 0.040	1.04 ± 0.22	2.06 ± 0.75	0.172 ± 0.042	0.46 ± 0.30	0.230	0.802	2.94	1.927
Sorocaba	Q1	0.221	0.379 ± 0.046	1.16 ± 0.25	2.16 ± 1.16	0.474 ± 0.097	1.47 ± 0.50	1.184	1.691	7.64	1.390
Sorocaba	Q2	0.153	0.391 ± 0.120	1.84 ± 0.77	1.44 ± 0.82	0.217 ± 0.038	0.54 ± 0.11	0.264	0.401	1.12	2.428
Sorocaba	Q3	0.220	0.419 ± 0.062	1.37 ± 0.27	3.21 ± 0.47	0.392 ± 0.034	1.15 ± 0.10	1.302	2.683	12.03	4.063
Sorocaba	Q4	0.135	0.227 ± 0.035	0.82 ± 0.12	1.54 ± 0.50	0.256 ± 0.038	1.12 ± 0.28	0.656	0.652	3.44	1.665
Mean of 16 cells			0.395	2.012	2.801	0.337	1.233	0.642	1.140	4.362	2.588

GNN-RF by at most 0.104 dB. Over all Q2–Q4 test nodes, the share within 2 km of the training blocks of the previous quadrant is 1.11% (109 745 of 9 884 081), so this proximity moves the blocked error of GNN-RF only slightly.

The comparison between the node-wise adaptation and GNN-RF is inconclusive relative to the pre-specified margin in both primary populations. Over all test nodes, the RSSI MAE differs by −0.0571 dB (95% CI [−0.1183, +0.0025], 80 paired runs; 0.3375 against 0.3946 dB), an interval that crosses the margin of ±0.045 dB. Over valid-target nodes, it differs by −0.7794 dB (95% CI [−1.8484, −0.1163]; 1.2327 against 2.0122 dB). This interval excludes zero on the side where the node-wise adaptation has the lower error but does not lie entirely beyond the margin of −0.35 dB, so the difference is also inconclusive.

Per cell, the node-wise adaptation has a lower mean RSSI MAE than GNN-RF in 12 of the 16 cells in both populations, with the same four exceptions: Bauru Q2, Campinas Q1, Sorocaba Q1, and Sorocaba Q4 (Table 6; all test nodes in Fig. 3). The four city means of the valid-target difference are all negative (Bauru −0.3116, Campinas −0.2504, Lins −2.3295, and Sorocaba −0.2263 dB). The largest in magnitude falls in Lins, whose test blocks differ in composition from those of the other cities.

The valid-target test nodes of Lins Q1 and Q2 are dominated by nodes at or above the −85 dBm coverage threshold. Of the valid-target test nodes, 93.2% in Lins Q1 (23 249 of 24 945) and 89.2% in Lins Q2 (109 989 of 123 306) have a target RSSI of at least −85 dBm, against 0 to 12.0% in the other 14 cells. On this population, over the seven cells with at least 1000 such nodes, the mean-of-cells RSSI MAE is 2.631 dB for the node-wise adaptation against 7.931 dB for GNN-RF, and the node-wise adaptation has the lower error in 34 of the 35 paired runs (Table 8).

The remaining contrasts of the node-wise adaptation are descriptive and carry no pre-specified margin. Valid-target path loss differs from that of GNN-RF by −1.0312 dB (95% CI [−1.9267, −0.1718]), whereas on sentinel nodes the RSSI MAE of the node-wise adaptation is the higher one, with a difference of 0.0162 dB (95% CI [+0.0012, +0.0299]). Against the graph-free control, the node-wise adaptation has a lower RSSI MAE, with differences of −0.3046 dB over all test nodes (95% CI [−0.5292, −0.0781]) and −1.5060 dB over valid-target nodes (95% CI [−2.4580, −0.5542]). On valid-target nodes, GNN-RF reaches 2.012 against 2.739 dB for the graph-free control, lower in 9 of the 16 cells. Both learners that receive the antenna-to-terrain relations of the graph, the node-wise adaptation and GNN-RF, thus have a lower error than the graph-free control. On valid-target nodes, the best calibrated empirical model of each cell (4.362 dB, Table 6) trails both.

C. MULTI-TARGET RECONSTRUCTION AND PHYSICS DECOMPOSITION

GNN-RF reconstructs the two operationally primary RF fields, total path loss and RSSI, directly in absolute decibels from a single multi-output head, with the valid-target errors given in Section V-B, and the same decoder emits a coverage probability. Because coverage in the reference field is RSSI thresholded at −85 dBm, this output is a derived classification rather than an independent target.

The remaining two channels, vegetation attenuation and

TABLE 7: Transfer chain versus training from scratch on the same blocked split (Section V-B). Each pair compares the quadrant model obtained by fine-tuning from the previous quadrant (chained: 20 epochs, learning rate 2×10^{-4}) with a model trained from scratch on the same data and split (35 epochs, 10^{-3}); test RSSI MAE in dB, all test nodes; Δ is chained minus scratch. The pairs form eight (city, seed) clusters ($n = 8$) nested in the four cities ($n = 4$).

City	Seed	Q	Scratch (dB)	Chained (dB)	Δ (dB)
Bauru	42	Q2	0.316	0.470	+0.154
Bauru	42	Q3	0.417	0.373	−0.044
Bauru	42	Q4	0.364	0.248	−0.116
Bauru	43	Q2	0.272	0.315	+0.044
Bauru	43	Q3	0.411	0.419	+0.008
Bauru	43	Q4	0.351	0.353	+0.002
Campinas	42	Q2	0.348	0.267	−0.081
Campinas	42	Q3	0.850	0.648	−0.202
Campinas	42	Q4	0.194	0.185	−0.009
Campinas	43	Q2	0.332	0.241	−0.091
Campinas	43	Q3	0.974	0.870	−0.105
Campinas	43	Q4	0.291	0.189	−0.102
Lins	42	Q2	0.317	0.496	+0.179
Lins	42	Q3	0.212	0.337	+0.125
Lins	42	Q4	0.271	0.261	−0.010
Lins	43	Q2	0.496	0.449	−0.048
Lins	43	Q3	0.253	0.230	−0.024
Lins	43	Q4	0.179	0.168	−0.011
Sorocaba	42	Q2	0.484	0.593	+0.109
Sorocaba	42	Q3	0.468	0.505	+0.038
Sorocaba	42	Q4	0.238	0.258	+0.020
Sorocaba	43	Q2	0.365	0.360	−0.004
Sorocaba	43	Q3	0.468	0.431	−0.038
Sorocaba	43	Q4	0.289	0.170	−0.119

TABLE 8: RSSI MAE of the node-wise adaptation and of GNN-RF on valid-target test nodes with target RSSI of at least −85 dBm, in the seven cells with at least 1000 such nodes (each cell the mean of five seeds; 35 paired runs). Nodes: number of test nodes of this population in the cell. The mean of cells is computed before rounding. The node-wise adaptation has the lower error in every cell. The population is described per cell, without an interval or a test.

Cell	Nodes	Node-wise adapt. MAE (dB)	GNN-RF MAE (dB)
Bauru Q1	7801	4.466	12.010
Bauru Q3	1151	0.572	2.115
Campinas Q1	6268	4.907	15.351
Campinas Q3	29 223	3.790	10.412
Lins Q1	23 249	3.566	10.049
Lins Q2	109 989	0.553	2.754
Sorocaba Q3	4505	0.562	2.823
Mean of cells		2.631	7.931

terrain diffraction, are supervised against the corresponding components of the generator, which are computed at a single 1800 MHz carrier and toward the nearest transmitter (documented in the Supplementary Material), and are reported as auxiliary outputs. Over valid-target nodes, GNN-RF reaches a mean-of-cells MAE of 1.11 dB on the vegetation channel and 0.83 dB on the diffraction channel. In the generator, the slope-based terrain term, local to each node, enters the total path loss, whereas the midpoint-obstacle knife-edge term, whose loss approximates the single knife-edge formulation of ITU-R P.526 [54], enters only the diffraction channel, computed toward the nearest transmitter.

The coverage class holds 1.38% of test nodes and 8.11% of valid-target nodes, and the probability output cut at 0.5 agrees with the predicted RSSI thresholded at −85 dBm on 99.61% of test nodes, pooled over cells and seeds (Cohen's κ=0.78). Against the label defined by the target RSSI, the thresholded predicted RSSI reaches an F1 of 0.547 and the probability output, cut at the fixed value of 0.5, an F1 of 0.441, as the mean of the nine cells that contain both classes.

D. FREE-SPACE-FLOOR ADMISSIBILITY

Free-space-floor admissibility is verified on the output rather than imposed: for each test node, the floor is set by the distance to the nearest transmitter at 1800 MHz. Of the 80 blocked runs, 69 have no test node below the floor, and across all runs 2536 of 65 930 590 test nodes fall below it, unevenly distributed by city (Table 9). This diagnostic is a bound test rather than the floor of the target: it takes the nearest transmitter at a fixed 1800 MHz carrier, whereas the generator serves each node from the strongest transmitter at that link's carrier.

With the serving transmitter's distance and carrier, 1.39×10^{-4} of valid-target test nodes fall below the free-space floor as the mean of cells (1.10×10^{-4} pooled), 1235 node-run violations out of 11 237 185 valid-target test-node evaluations, all from 9 of the 80 runs in four cells (Bauru Q1, Campinas Q1, Campinas Q3, and Lins Q1). A constant predictor, the median of the valid targets outside the test partition, never falls below the floor in any of the 16 cells (minimum margin 29.7 dB), so passing the floor does not separate a physical field from a degenerate one. The NDVI term of Eq. (18) is a sign condition on a correlation rather than a per-node relation, so no per-node vegetation fraction is reported.

E. ROBUSTNESS AND ABLATION

City means rise from 0.348 dB in Lins through 0.354 dB in Sorocaba and 0.407 dB in Bauru to 0.470 dB in Campinas, and across the 80 blocked runs variation between cells dominates variation between seeds (Section IV-E; Fig. 3). Against this variation, Table 10 and Fig. 4 report how much test RSSI MAE degrades when the graph of a trained model is damaged, paired by cell over the eight Lins and Bauru cells. Shuffling antenna-to-terrain edges adds 0.18 dB, shuffling terrain-to-terrain edges 0.67 dB, shuffling both 0.80 dB, and removing all edges 1.72 dB. Terrain-to-terrain connectivity matters more than antenna-to-terrain connectivity in seven of the eight cells, the exception being Bauru Q2, and in Lins Q1 the empty graph barely changes the error (0.132 dB). Only the direction of the difference between arms is reported, since eight cells leave their ratio too uncertain to quote.

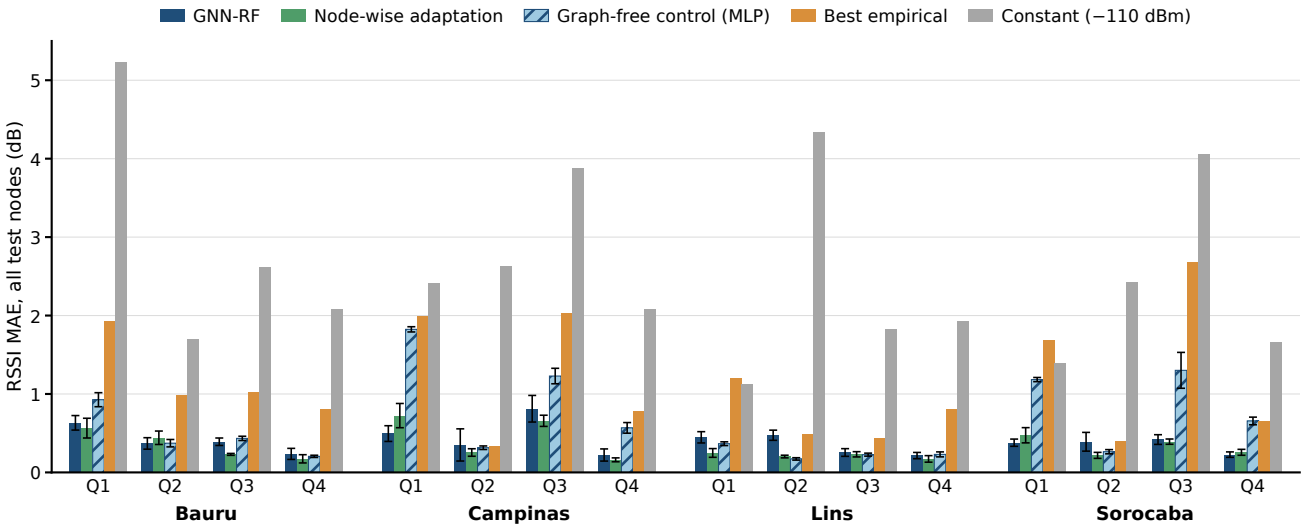


FIGURE 3: RSSI MAE on all test nodes of the spatially blocked partition, 16 city–quadrant cells. GNN-RF is lower than the best calibrated empirical model in 15 of 16 cells and than the graph-free control in 10. The node-wise adaptation has a lower MAE than GNN-RF in 12 of 16 cells. Bars: mean of five seeds; whiskers: ± s.d. Values are in Table 6.

TABLE 9: Free-space-floor admissibility of GNN-RF over the 80 blocked runs, all test nodes, with the floor at the distance to the nearest transmitter and 1800 MHz (Section V-D). Node counts are recovered from the recorded per-run fraction and number of test nodes. The last block repeats the test on the valid-target nodes with the distance and carrier of the transmitter that serves each node.

City	Runs with no node below the floor	Nodes below the floor	Test nodes evaluated
Bauru	18 of 20	115	16 458 440
Campinas	13 of 20	2118	16 438 370
Lins	18 of 20	303	16 582 940
Sorocaba	20 of 20	0	16 450 840
Total	69 of 80	2536	65 930 590
Serving transmitter, valid-target nodes			
Total	71 of 80	1235	11 237 185

The graph-free control (Section IV-D) covers all four cities rather than the two cities of the ablation above. At the city unit (n=4), the primary analysis, GNN-RF has lower mean RSSI MAE than the graph-free control by $0.247\,\mathrm{dB}$, but the 95% CI crosses zero (p=0.205); with only four cities, no two-sided non-parametric test on this design can reach $p < 0.125$, a bound that applies to any per-city result quoted from this control.

At the finer cell unit (n=16), the mean difference, GNN-RF minus the control, is unchanged and its significance is sensitive to the inferential method (95% CI [−0.482, −0.013], p=0.0397 paired t, p=0.117 Wilcoxon); the same direction and magnitude hold at the chain unit (n=20, p=0.00103). The chain unit shares a Q1 checkpoint and spatial adjacency across its four cells, so it does not add an independent estimate of the effect beyond the cell-level reading above.

The sign of the advantage survives every leave-one-city-out window, but its size does not: excluding Campinas or Sorocaba roughly halves the mean effect (-0.159 and $-0.164\,\mathrm{dB}$, against -0.363 excluding Lins and -0.304 excluding Bauru). This pattern implies that Lins's own contribution to the aggregate runs opposite in sign to the other three cities. Across the 16 cells, the advantage correlates moderately with the fraction of valid-target test nodes ($r = -0.56$), an association that is exploratory and does not establish a causal relationship.

Retraining GNN-RF without any penalty term (C1) is inconclusive relative to the pre-specified margin in both primary populations. Over all test nodes the RSSI MAE differs by $-0.0593\,\mathrm{dB}$ (95% CI [−0.1249, +0.0156], 80 paired runs), and over valid-target nodes by $0.1497\,\mathrm{dB}$ (95% CI [−0.3082, +0.8106]). Lins dominates the valid-target interval, with a city mean of $+0.9916\,\mathrm{dB}$ against -0.2669 to $-0.0026\,\mathrm{dB}$ in the other three cities. Without any penalty, the same architecture still keeps a mean error well below that of the best calibrated empirical model of each cell, 0.335 against 1.140 dB over all test nodes and 2.162 against 4.362 dB over valid-target nodes.

Removing only the monotonicity term (C2) is also inconclusive, although its all-node interval excludes zero. Over all test nodes the difference is $-0.0865\,\mathrm{dB}$ (95% CI [−0.1323, −0.0391]), an interval that crosses the margin of $-0.045\,\mathrm{dB}$, and over valid-target nodes it is $-0.0768\,\mathrm{dB}$ (95% CI [−0.3512, +0.3093]). Sentinel nodes account for a fraction of 0.817 of the all-node difference, under an exact per-run decomposition of the all-node MAE into its valid-target and

sentinel parts defined after the runs.

The bias of C2, the mean signed error of prediction minus target, is inconclusive on valid-target nodes against its margin of ± 0.1 dB, set before the runs. There it changes by $+0.1174$ dB for RSSI (95% CI $[-0.4070, +0.8791]$) and by -0.4822 dB for path loss (95% CI $[-1.4799, +0.7790]$), and both intervals cross that margin. On sentinel nodes, where the same ± 0.1 dB serves as a criterion set after the runs, the mean signed RSSI error shifts by $+0.0245$ dB (95% CI $[+0.0049, +0.0412]$) and stays inside it.

The physical admissibility of the outputs of C1 and C2 differs from that of the reference runs only within intervals that contain zero. Table 11 reports two quantities per model: the fraction of valid-target nodes predicted below the free-space floor of their serving transmitter, and the fraction of same-transmitter pairs in which the farther node is predicted more than 1 dB stronger than the nearer one. For GNN-RF, the node-wise adaptation, and the graph-free control, none of the predicted floor violations falls on a node where the target itself lies below the floor.

F. SPATIAL ERROR ANALYSIS

Test-set RSSI MAE decomposes very differently between the two populations that make up each cell. In all four cities, nodes with a valid path-loss target concentrate most of the absolute error, while sentinel nodes (-110 dBm) carry near-zero MAE: valid-target MAE ranges from 1.296 dB (Sorocaba) to 3.571 dB (Lins) as city means over four cells and five seeds, against 0.134 dB for the sentinel population as the mean of cells, and the valid-target population accounts for 71.89% of total absolute error, pooled over cells and seeds.

The blocked split's geometry bounds how far a test node can be from training data. Across the four cities, no test node lies farther than 8.58 km from the nearest training block edge, and the minimum distance falls between 1.963 and 1.971 km, consistent with the pre-specified 2 km exclusion buffer. Of the 20 test blocks the split lays out per city, only 17 (Bauru, Campinas, Lins) or 15 (Sorocaba) contain a node in the released dataset, because the last 10 km grid strip covers only about 0.97 km of the tile's extent; the rest of the nominal 20 blocks are geometrically empty, not missing data.

Error is spatially concentrated within the test area (Fig. 5, signed RSSI error of the seed-42 models on the Q1 test nodes of each city). Half of the total absolute RSSI error, all test nodes included, falls within a fraction of the tile that ranges from 1.5% in Lins to 10.0% in Sorocaba, wider (5.3% to 27.5%) for valid-target nodes. Comparing near-training (< 3 km) against far (≥ 5 km) valid-target MAE within the same test block, 8 of the 10 blocks large enough to pair (three in Bauru, four in Campinas, three in Sorocaba; Lins has none with enough valid-target nodes on both sides) show lower error near the training boundary, a border-optimism signal with a median paired ratio of 0.872. The city-pooled ratio inverts this signal in Bauru and Campinas, a Simpson's-paradox risk anticipated before the result, so the evidence used here is the paired-by-block reading, not the city pool.

The spatial correlation of the valid-target residual at short lag bins is well below that of the target field itself: at 1–2 km it is 0.53 in Bauru and 0.58 in Lins, against 0.93 and 0.90 for the reference field at the same bin, indicating that most of the target's long-range structure is already explained rather than left in the residual. The formal range estimate of the residual saturates at the 50 km grid ceiling in Bauru and Campinas, a sign of instability from few pairs at long distances, not of real structure, and is not reported as a single comparable range.

G. UNCERTAINTY UNDER SPATIALLY BLOCKED EVALUATION

Epistemic dispersion is estimated by Monte Carlo Dropout [16] with $T=50$ stochastic passes on the 16 seed-42 checkpoints, activating only the dropout layers, attention dropout in the GATv2 layers included. Intervals come from the MC percentiles at five nominal levels, from a split-conformal correction calibrated on the validation blocks of each cell, and, as a baseline, from the validation residuals of the deterministic model (Table 12, Fig. 6). On valid-target test nodes, MC Dropout intervals contain the target in 0.369, 0.592, 0.683, 0.738, and 0.801 of cases at the 50, 80, 90, 95, and 99% levels.

The conformal correction reaches 0.610 to 0.951 on average, but its 95% coverage ranges from 0.26 to 1.00 across cells, with 7 of 16 cells more than 0.05 from nominal, and the deterministic-residual baseline covers as much (0.659 to 0.942). No method meets the criterion of Section IV-E: MC Dropout qualifies in 3 of 16 cells, the conformal correction in 1, and the residual baseline in 2.

The MC-mean prediction has valid-target MAE of 2.35 dB against 2.17 dB for the deterministic prediction, with mean MC spread of 1.67 dB and CRPS of 1.86 dB. Under spatial blocking, the dropout output is therefore a measure of in-domain dispersion rather than calibrated uncertainty: validation and test blocks are not exchangeable, so a quantile calibrated on one set of blocks does not transfer to another, and a residual-based interval achieves the same coverage without stochastic passes.

H. CROSS-CITY TRANSFER

Lins-trained seed-42 checkpoints are applied without fine-tuning to the blocked test partitions of Lins, Campinas, and Sorocaba (Table 13); zeroing the topographic features raises mean RSSI MAE in every city tested, in-domain and out-of-domain alike, from 0.399 dB to 2.757 dB in Lins ($+591.2\%$), from 1.353 dB to 3.304 dB in Campinas ($+144.1\%$), and from 1.622 dB to 3.931 dB in Sorocaba ($+142.4\%$), with the percentage taken over the full-terrain arm. The same pattern holds at larger absolute magnitude on valid-target nodes for both RSSI and path loss, and the in-domain increase remains far larger than either out-of-domain city in both populations.

The reversal obtained in the preliminary experiment (-8 to -9% out-of-domain) came from a different evaluation: it scored the entire 12.96-million-node graph of each quadrant, training and validation nodes included, rather than the held-

TABLE 10: Structural ablation on the spatially blocked test split (all test nodes). The four edge-ablation rows use Lins and Bauru (Q1–Q4, seed 42 checkpoints; $n = 8$ cells, three shuffling seeds per arm); entries are the change in test RSSI MAE relative to the trained model with the real graph, paired by cell; positive means worse. The last three rows are a different contrast: the paired difference GNN-RF minus the graph-free control (Section IV-D), in all four cities with five training seeds per cell, tested in the three pre-specified units (city; city–quadrant cell; seed chain, the mean of the four quadrants of one seed); negative means GNN-RF is more accurate. Cells and chains are not independent, since Q2–Q4 inherit the Q1 checkpoint of the same seed.

Arm	ΔMAE (dB)	95% CI	p (paired t)	p (Wilcoxon)
Antenna→terrain edges shuffled (degree preserved)	+0.181	[+0.019; +0.343]	0.0332	0.0391
Terrain→terrain edges shuffled (degree preserved)	+0.673	[+0.262; +1.084]	0.0061	0.0078
Both relations shuffled	+0.798	[+0.428; +1.169]	0.0014	0.0078
Empty graph (no message passing)	+1.723	[+0.773; +2.674]	0.0036	0.0078
GNN-RF minus graph-free control (MLP, 1,812,515 nominal params), four cities				
unit: city ($n = 4$)	−0.247	[−0.736; +0.241]	0.2051	0.3750
unit: cell ($n = 16$)	−0.247	[−0.482; −0.013]	0.0397	0.1167
unit: seed chain ($n = 20$)	−0.247	[−0.381; −0.114]	0.0010	0.0032

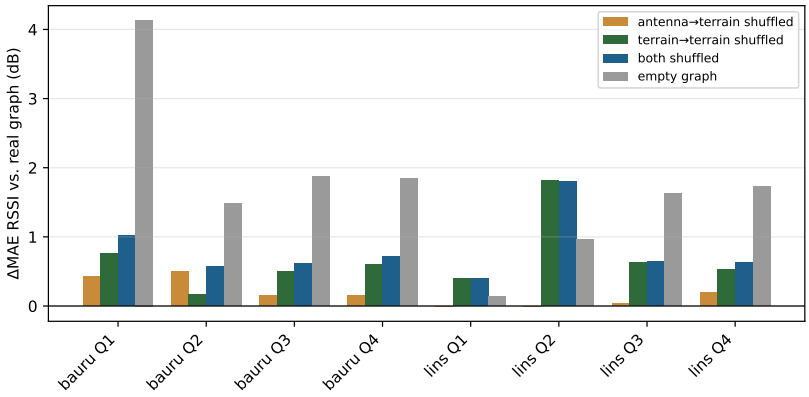


FIGURE 4: Structural ablation on the spatially blocked test split: change in RSSI MAE per cell from shuffling one edge relation, both relations, or removing all edges, relative to the trained model on the real graph (Lins and Bauru, Q1–Q4, seed-42 checkpoints).

out blocked test partition, which covers only about 6.3–6.4% of that graph. Under the criterion fixed before this result, the blocked measurement confirms the qualitative pattern expected of a terrain-dependent model: the topographic advantage is positive everywhere and larger in-domain than out. The blocked measurement contradicts the earlier number, not this pattern, and the two are not comparable value by value.

I. COMPUTATIONAL COST

Between GNN-RF and the graph-free control, the cost difference shows up as wall-clock time and memory. On Bauru, on the same GPU, GNN-RF trains in 3987 s from scratch (Q1, 35 epochs) and 2323 s under the transfer schedule (Q4, 20 epochs), with peak VRAM of 7475 and 7466 MB, against 616 s and 409 s, with 653 and 676 MB, for the graph-free control (Table 14); these figures come from a single cell and a single seed. The accuracy gain over the graph-free control (Section V-E) therefore comes at several times the training time and close to an order of magnitude more peak memory.

VI. DISCUSSION

A. INTERPRETATION OF RESULTS

The structural ablation and the graph-free control bound what the terrain graph contributes on this reference, over two different scopes. In the eight Lins and Bauru cells of the ablation the trained model relies on its graph: damaging terrain-to-terrain edges costs more than damaging antenna-to-terrain edges in seven of the eight cells, and removing all edges costs accuracy except in Lins Q1, where the empty graph barely changes the error (Section V-E).

Across all four cities, GNN-RF is ahead of the graph-free control, but its city-level interval crosses zero and the cell-level result is sensitive to the inferential method (Section V-E). In Bauru Q1, the cell with the largest valid-target fraction (0.328), GNN-RF is ahead of the graph-free control in every training seed, an association between valid-target fraction and graph advantage that is exploratory, without an inferential test. This sample does not support generalizing a graph advantage to measured fields.

The node-wise adaptation receives the antenna-to-terrain relations of the graph and no terrain-to-terrain message, and

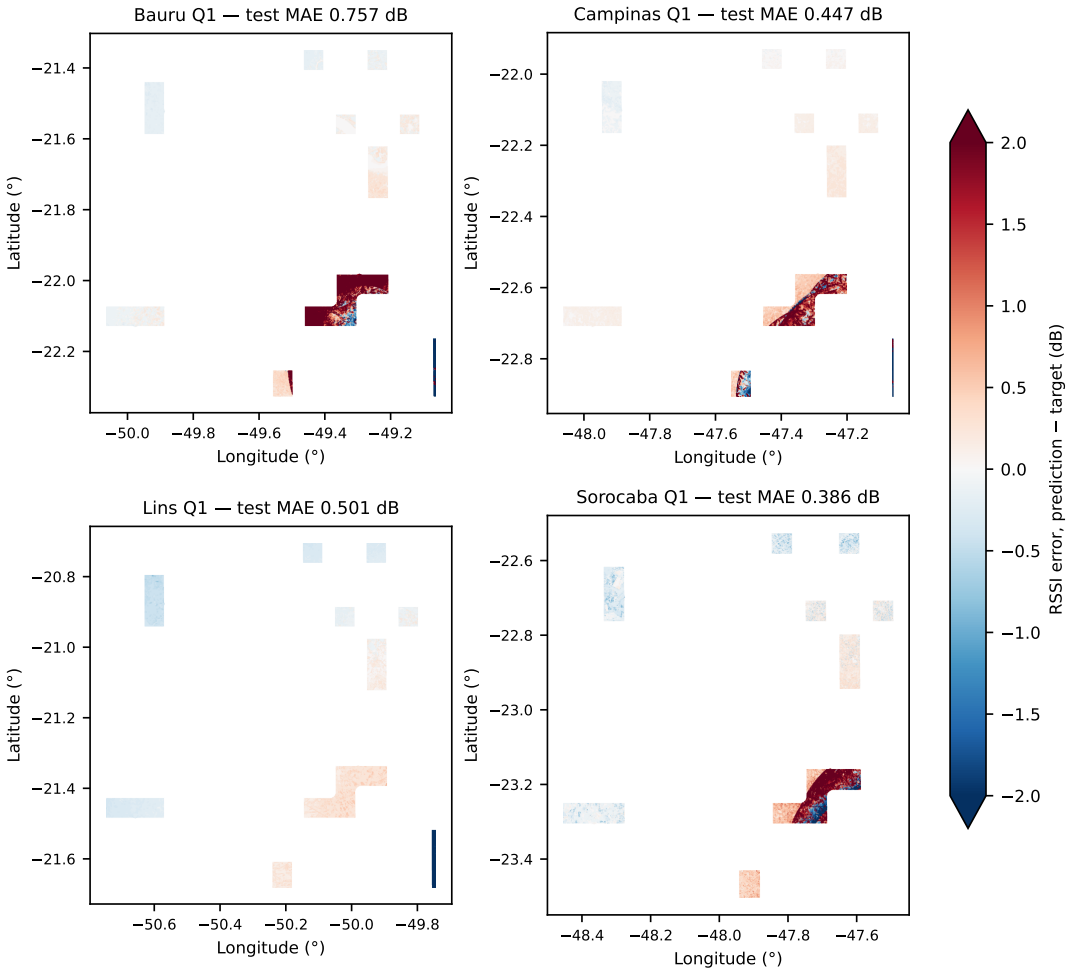


FIGURE 5: Signed RSSI prediction error (prediction − target, dB, color scale clipped at ±2 dB) on the spatially blocked test nodes of Q1 in each city (seed 42); the rectangles are the test blocks, whose layout repeats across cities because every partition uses the same block grid and seed. Axes in degrees of longitude and latitude.

with about one fifth of the parameters it has a lower valid-target RSSI MAE than the graph-free control (Section V-B). Over the same nodes, its difference from GNN-RF is inconclusive relative to the pre-specified margin, so terrain-to-terrain message passing yields no detectable gain in accuracy over what the node-wise adaptation already achieves. The structural ablation measures the cost of damaging the graph of a model trained with it, and the node-wise adaptation measures whether terrain-to-terrain messages are needed for accuracy.

On measured 4G received power with whole sites held out [28], the GNN whose pixel edges follow lines from the transmitter has the lowest RMSE in both reported test sets, 9.8 dB on the raw set and 8.5 dB on the indoor-filtered set, against 10.2 and 9.1 dB for gradient boosting on node features, whereas grid edges alone give 10.5 dB on the raw set, behind the gradient-boosting model, and 8.9 dB on the indoor-filtered set, ahead of it. On measured data, adding edges aligned with the transmitter to the grid edges gives a lower error than a model that sees only the node, in the direction of the antenna-to-terrain result reported here, although the edges of that model join pixels to pixels and each model has a single reported value.

Next to these learned controls, the comparison with empirical models survives calibration. Paired against the best calibrated model of each cell, GNN-RF has the lower error over all test nodes in 15 of the 16 cells (Section V-B); over valid-target nodes the mean margin widens, and the four cells where GNN-RF trails are Campinas Q2, Lins Q1, Lins Q2, and Sorocaba Q2.

On strong-signal nodes near the serving transmitter, GNN-RF and the graph-free control underestimate RSSI. On valid-target nodes with target RSSI of at least −70 dBm, their node-weighted bias is −13.90 and −11.93 dB, respectively, against −2.73 dB for the node-wise adaptation; on those within 250 m of the transmitter, it is −30.96 and −28.88 dB, respectively, against −14.15 dB. Within 2 km of the transmitter, the least-squares slope of path loss against $\log_{10}$ of

TABLE 11: Physical admissibility of the outputs under the blocked protocol, for the two quantities of Section V-E. Floor: mean of 80 runs per model. Monotonicity: pairs of valid-target nodes whose distances to their shared serving transmitter differ by a factor of at least 2, mean of 40 runs per model over the 8 of 16 cells that have such pairs. The reference-field row scores the target itself (floor: mean of the 16 cells; monotonicity: mean of the 8 cells, with the maximum of a single cell in parentheses). The 95% intervals of the paired differences of C1 and C2 against the reference contain zero for both quantities.

Model	Below floor (fraction, 80 runs)	Monotonicity violations (fraction, 40 runs)
GNN-RF (reference)	1.39×10^{-4}	0.0446
No-penalty retraining (C1)	9.21×10^{-5}	0.0378
No-monotonicity retraining (C2)	1.14×10^{-5}	0.0304
Node-wise adaptation	2.04×10^{-5}	0.0017
Graph-free control	0	0.0252
Reference field	2.43×10^{-4}	4.14×10^{-4} (0.0033)

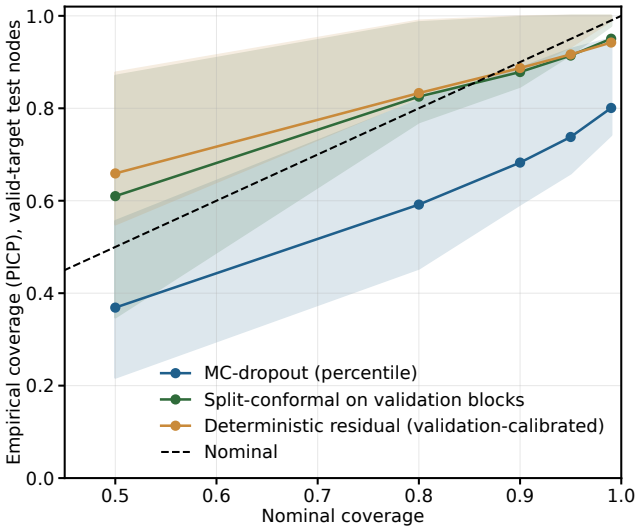


FIGURE 6: Prediction-interval coverage against nominal level under the spatially blocked protocol, by method, mean over the 16 cells with interquartile band.

distance is 8.05 dB per decade for GNN-RF and for the graph-free control and 23.22 dB per decade for the node-wise adaptation, against 34.95 dB per decade for the target, each a mean over the four cells with at least 50 such valid-target nodes (Bauru Q1, Campinas Q1, Campinas Q3, and Lins Q1). The reconstruction near the serving transmitter is the tightest limit of this study.

GNN-RF, the node-wise adaptation, and the graph-free control, the three networks sharing the output decoder, and GNN-RF retrained without any penalty term all place a smaller fraction of valid-target nodes below the free-space floor than the target itself (Table 11). The monotonicity violation rate, in contrast, varies among networks trained with the same loss and decoder, so it is not set by them, and retraining without the penalty terms changes it only within an interval that contains zero. GNN-RF keeps its floor admissibility high but uneven across cities, clean in every Sorocaba run and in most Bauru and Lins runs, with more exceptions in Campinas.

The claims of this work rest on the blocked evaluation itself, on the joint reconstruction of path loss and RSSI in absolute decibels, and on controls that make the contribution of the graph measurable; the uncertainty evaluation adds a negative result, since no interval method tested reaches nominal per-cell coverage under spatial blocking. Free-space-floor admissibility is common to every network trained here on this target and satisfied almost everywhere by the target itself: it is verified on the outputs rather than credited to the graph.

Because the reference field is fully specified, the bench also measures the evaluation itself: the nearest-neighbor contrast of Section V-A, in line with the optimistic estimates that random cross-validation gives for spatially clustered data [55], separates spatial learning from partition leakage independently of where the labels come from, so the design can be reused unchanged with measured RSSI or drive-test path loss. The native 30 m resolution of the Copernicus DEM (Section III-A), however, bounds the achievable spatial granularity of the RF predictions.

B. METHODOLOGICAL IMPLICATIONS FOR RF EMULATION

Within the scope of deterministic RF field emulation, the results suggest that sequential fine-tuning saves compute: continuing from an adjacent checkpoint for 20 epochs shows no systematic difference from training from scratch for 35 epochs (Section V-B). They also suggest that, for surrogate emulation pipelines, terrain feature engineering weighs more than the one architectural contrast measured here, GNN-RF against a graph-free control, since zero-shot checkpoints transferred to unseen cities, tested on a single seed, suffer more from ablating topography than from the domain shift itself (Section V-H). Neither takeaway prescribes physical network deployment, which remains conditional on empirical field validation as detailed below.

C. SIMULATION-DERIVED TARGETS AND VALIDITY BOUNDS

A fundamental constraint of this study is that all target labels come from the deterministic propagation generator of Section IV, which computes path loss, RSSI, and coverage from antenna parameters, node-level terrain and vegetation features, and empirical correction factors without tracing terrain profiles. The reported MAE values (0.395 dB on all blocked test nodes) therefore represent reconstruction fidelity relative to these simulation outputs, not prediction accuracy against physical RF measurements.

The generator also omits mechanisms that a full link-level simulation would include: diffraction over real obstacles,

TABLE 12: Prediction-interval coverage on the spatially blocked test split, valid-target nodes, RSSI channel, 16 cells (seed-42 checkpoints, 50 stochastic passes). PICP is the mean over cells with the [min; max] across cells; width is the mean interval width. The last column counts cells within ± 0.05 of nominal simultaneously at 80, 90 and 95 % (calibration criterion in Section IV-E).

Method		Nominal level					Cells within 0.05
		50 %	80 %	90 %	95 %	99 %	(80/90/95)
MC-dropout (percentile)	PICP	0.369 [0.01; 0.67]	0.592 [0.02; 0.92]	0.683 [0.03; 0.97]	0.738 [0.06; 0.98]	0.801 [0.15; 0.99]	3/16
	width (dB)	2.0	4.0	5.2	6.1	7.2	
MC-dropout (Gaussian)	PICP	0.383 [0.01; 0.70]	0.616 [0.02; 0.94]	0.712 [0.03; 0.98]	0.776 [0.09; 0.99]	0.869 [0.31; 1.00]	3/16
	width (dB)	2.2	4.3	5.5	6.5	8.6	
Split-conformal on validation blocks	PICP	0.610 [0.01; 1.00]	0.825 [0.06; 1.00]	0.878 [0.16; 1.00]	0.914 [0.26; 1.00]	0.951 [0.38; 1.00]	1/16
	width (dB)	6.0	10.8	14.0	17.6	25.0	
Deterministic residual (validation-calibrated)	PICP	0.659 [0.01; 1.00]	0.833 [0.05; 1.00]	0.887 [0.11; 1.00]	0.917 [0.15; 1.00]	0.942 [0.20; 1.00]	2/16
	width (dB)	5.5	11.3	16.5	22.7	34.9	

TABLE 13: Cross-city transfer under the spatially blocked protocol (Section V-H): Lins checkpoints (seed 42, one per quadrant) on the blocked test partition of the same quadrant in Lins (in-domain), Campinas and Sorocaba. The no_topo arm zeroes the eight topographic input features of the terrain nodes at inference. Entries are unweighted mean MAE over Q1–Q4 for all nodes (sentinel included), valid-target nodes for RSSI, and valid-target nodes for path loss (PL); Δ is no_topo minus full, and the change is Δ divided by the full-arm MAE.

		RSSI, all nodes			RSSI, valid-target			PL, valid-target		
Target city	Arm	MAE (dB)	Δ (dB)	Change	MAE (dB)	Δ (dB)	Change	MAE (dB)	Δ (dB)	Change
Lins (in-domain)	full	0.399	–	–	3.92	–	–	3.64	–	–
Lins (in-domain)	no_topo	2.757	+2.358	+591.2%	23.51	+19.59	+500.4%	29.98	+26.34	+722.6%
Campinas (OOD)	full	1.353	–	–	5.73	–	–	7.54	–	–
Campinas (OOD)	no_topo	3.304	+1.950	+144.1%	17.54	+11.81	+206.1%	25.02	+17.48	+231.9%
Sorocaba (OOD)	full	1.622	–	–	7.50	–	–	8.72	–	–
Sorocaba (OOD)	no_topo	3.931	+2.309	+142.4%	20.75	+13.26	+176.9%	27.45	+18.72	+214.7%

TABLE 14: Computational cost in the spatially blocked campaign (Bauru, seed 42): GNN-RF and the graph-free control (MLP, Section IV-D). Q1 trains from scratch (35 epochs); Q4 transfers (20 epochs). Size assumes 4 bytes per parameter and is given in MiB (2^{20} bytes); peak VRAM is the maximum memory allocated by PyTorch during training and is given in MB (10^6 bytes).

			Time (s)		Peak VRAM (MB)	
Model	Params	Size (MiB)	Q1	Q4	Q1	Q4
GNN-RF	1 812 515	6.91	3987	2323	7475	7466
MLP	1 812 515	6.91	616	409	653	676

Fresnel-zone clearance, line-of-sight testing, and multipath or ground-reflection effects are absent, and the knife-edge channel uses a single midpoint obstacle rather than the profile maximum. Its two-term link budget carries no noise floor, receiver noise figure, interference from neighboring cells, or SINR calculation, and coverage adds no fading margin, outage probability, shadowing standard deviation, or receiver sensitivity, remaining a hard threshold on the link budget rather than a link-quality decision. The generator draws no random term, so no log-normal shadowing is present; the uniform slope draw of the early pipeline was replaced by the DEM slope.

Both empirical base-loss models also run partly outside their validated range. The accuracy figures therefore bound fidelity to this extrapolated field, not agreement with Okumura–Hata or COST 231 inside their original envelopes.

The gap between simulated and measured propagation in the Brazilian context is not quantified in this work, and studies that compare sub-GHz path-loss predictions with field measurements, in Cartagena, Spain [10] and in the Amazon [9], report errors larger than the sub-decibel reconstruction figures obtained here. Validation against even partial field measurement data would be required before any operational use.

All evaluations are restricted to four cities in São Paulo state (Brazil), and generalization to other biomes, frequency bands, or network topologies has not been tested. The edge ablation covers only Lins and Bauru, whereas the graph-free control spans all four cities; the uncertainty evaluation covers all 16 cells but is limited to the seed-42 checkpoints, each sampled with 50 Monte Carlo passes.

The 2 km exclusion buffer is narrower than the spatial autocorrelation range [56] of the reference field, measured at 20 km in Lins and 30 km in Bauru and not measured in the other two cities, so the blocked results retain some dependence between training and test. Buffering against the full 20–30 km range is geometrically infeasible within quadrant-scale tiles. The 2 km buffer suppresses immediate mesh-scale interpolation, as the order-of-magnitude error inflation of the nearest-neighbor predictor in Section V-A reflects, but residual regional autocorrelation remains an explicit bound

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on claim independence.

In three Sorocaba cells, 0.03–0.06% of nodes with a valid target carry an RSSI below −150 dBm that the target post-processing did not impute, and correcting Q3, the only one of these cells where it changes test error, in both arms does not change the conclusion of the comparison against the graph-free control in Section V-E.

D. LIMITATIONS AND FUTURE DIRECTIONS

The blocked cross-city results, in which zeroing topographic features hurts accuracy everywhere and more inside the training domain than outside it, motivate domain adaptation that preserves that terrain dependence when transferring across cities without full fine-tuning. Transfer of a radio-map model between indoor environments has relied on fine-tuning the whole network [50], whereas unsupervised domain adaptation guided by geographic coordinates already serves remote-sensing segmentation [57]. Interval methods that account for the non-exchangeability of spatial blocks are also needed, since split-conformal calibration on separate blocks did not transfer here (Table 12) and conformal coverage is no longer guaranteed once exchangeability is violated [38].

VII. CONCLUSIONS

This paper evaluates GNN-RF under a spatially blocked protocol because partition choice changes what a radio-map estimator's accuracy can support: a nearest-neighbor copy that fails under blocking looks adequate under a random node split in all 16 city–quadrant cells, and a geographic-split control on a RadioUNet adaptation from a preliminary experiment (Section V-A) failed in half of the same cells.

Under that blocked protocol, across all 16 cells (80 runs), GNN-RF reconstructs path loss and RSSI in absolute decibels, and coverage, over large-scale terrain graphs (12.96×10^6 nodes), with a test RSSI MAE of $0.395\,\mathrm{dB}$ (95% CI $[0.309, 0.480]$, all test nodes). Paired against the best calibrated empirical model of each cell, it wins by a mean of $0.746\,\mathrm{dB}$ ($p{=}2.59 \times 10^{-4}$) in 15 of the 16 cells and loses by $0.009\,\mathrm{dB}$ in Campinas Q2, and the margin widens on valid-target nodes. Its outputs stay within their activation ranges, and the fraction of valid-target nodes whose predicted path loss falls below the free-space floor of the serving transmitter is 1.39×10^{-4}, below the 2.43×10^{-4} of the target itself.

The accuracy advantage over the graph-free control follows the antenna-to-terrain relations of the graph. On valid-target nodes, the two learners that receive the antenna-to-terrain relations of the graph, GNN-RF and the node-wise adaptation of RadioUNet, reach RSSI MAE of 2.0122 and 1.2327 dB, below both the graph-free control matched in nominal parameters (2.7387 dB) and the best calibrated empirical model of each cell (4.362 dB). Damaging the graph degrades the trained model on the eight Lins and Bauru cells, by $1.72\,\mathrm{dB}$ on average when all edges are removed, except in Lins Q1, where the empty graph leaves the error nearly unchanged.

Neither the terrain-to-terrain messages nor the penalty terms account for a resolvable part of that accuracy: the difference between the node-wise adaptation and GNN-RF is inconclusive relative to the pre-specified margin, and so are the differences of GNN-RF retrained without all four penalty terms or without the monotonicity term alone. Free-space-floor admissibility is common to every network trained here on this target and satisfied almost everywhere by the target itself, so it is not a contribution of the graph.

Training each quadrant from scratch shows no systematic advantage for the transfer chain, and under the same blocked protocol, zeroing topographic features raises cross-city error everywhere, more inside the training domain than outside it. The uncertainty evaluation is a negative result: under spatial blocking, neither MC Dropout nor its split-conformal correction reaches nominal coverage per cell, and a residual-based interval covers as much at no extra cost.

Future work will prioritize validation against field RF measurements to quantify the gap between simulation-derived reconstruction accuracy and physical-world prediction error. A wider exclusion buffer, non-sequential transfer strategies, and other frequency bands and regions beyond São Paulo state would extend the evaluation, and coupling DEM super-resolution, for which learned methods already exist [58], [59], with RF reconstruction in end-to-end pipelines is a further direction that the released benchmark makes testable.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest regarding this work.

DATA AVAILABILITY

The target generator and the target corrector, identified by SHA-256 digest in the Supplementary Material, the 16×3 partition masks with their hashes and a verifier, the evaluation and baseline scripts, the training scripts, and 400 run logs, 400 checkpoints, and 400 prediction files, 80 each for GNN-RF, the graph-free control, the two retrained arms, and the node-wise adaptation, are available at https://doi.org/10.5281/zenodo.23601078. The 16 reference fields are not in the deposit: they are identified by SHA-256 digest and can be regenerated with the released generator and corrector and the public inputs. The complete generator specification is published as Supplementary Material.

REFERENCES

[1] M. Hata, "Empirical formula for propagation loss in land mobile radio services," IEEE Transactions on Vehicular Technology, vol. 29, no. 3, pp. 317–325, 1980.

[2] E. Damosso and L. M. Correia, Eds., Digital Mobile Radio Towards Future Generation Systems: COST 231 Final Report. Luxembourg: Office for Official Publications of the European Communities, 1999, EUR 18957, ISBN 92-828-5416-7. [Online]. Available: https://op.europa.eu/en/publication-detail/-/publication/f2f42003-4028-4496-af95-beaa38fd475f

[3] M. Vasudevan and M. Yuksel, "Machine learning for radio propagation modeling: A comprehensive survey," IEEE Open Journal of the Communications Society, vol. 5, pp. 5123–5153, 2024.

[4] ITU-R, "Recommendation ITU-R P.1546-6: Method for point-to-area predictions for terrestrial services in the frequency range 30 MHz to 4000 MHz," International Telecommunication Union, Geneva, Switzerland, Tech. Rep., 2019. [Online]. Available: https://www.itu.int/rec/R-REC-P.1546-6-201908-I/en

[5] 3GPP, "5G; study on channel model for frequencies from 0.5 to 100 GHz (3GPP TR 38.901 version 16.1.0 Release 16)," ETSI, Sophia Antipolis, France, Tech. Rep. ETSI TR 138 901 V16.1.0, Nov. 2020. [Online]. Available: https://www.etsi.org/deliver/etsi_tr/138900_138999/138901/16.01.00_60/tr_138901v160100p.pdf

[6] J. Thrane, D. Zibar, and H. L. Christiansen, "Model-aided deep learning method for path loss prediction in mobile communication systems at 2.6 GHz," IEEE Access, vol. 8, pp. 7925–7936, 2020.

[7] Ç. Yapar, F. Jaensch, R. Levie, G. Kutyniok, and G. Caire, "Overview of the first pathloss radio map prediction challenge," IEEE Open Journal of Signal Processing, vol. 5, pp. 948–963, 2024.

[8] J. C. Silva, G. L. Siqueira, and P. V. G. Castellanos, "Propagation model for path loss through vegetated environments at 700–800 MHz band," Journal of Microwaves, Optoelectronics and Electromagnetic Applications, vol. 17, no. 1, pp. 179–187, 2018.

[9] W. G. Lima, A. V. R. Lopes, C. M. M. Cardoso, J. P. L. Araújo, M. C. A. Neto, M. E. L. Tostes, A. A. Nascimento, M. Rodriguez, and F. J. B. Barros, "LoRa technology propagation models for IoT network planning in the Amazon regions," Sensors, vol. 24, no. 5, p. 1621, 2024.

[10] J. Lorente-López, I. Rodríguez-Rodríguez, J.-V. Rodríguez, J. M. Molina García-Pardo, and J. Aznar-Poveda, "Machine-learning-based urban path loss prediction at 900 MHz: Principal component analysis, clustering, feature importance and regression," IEEE Open Journal of Antennas and Propagation, vol. 7, no. 1, pp. 40–56, 2026.

[11] R. Levie, Ç. Yapar, G. Kutyniok, and G. Caire, "RadioUNet: Fast radio map estimation with convolutional neural networks," IEEE Transactions on Wireless Communications, vol. 20, no. 6, pp. 4001–4015, 2021.

[12] S. Zhang, A. Wijesinghe, and Z. Ding, "RME-GAN: A learning framework for radio map estimation based on conditional generative adversarial network," IEEE Internet of Things Journal, vol. 10, no. 20, pp. 18 016–18 027, 2023.

[13] A. Shibli and T. Zanouda, "Data-driven radio environment map estimation using graph neural networks," in 2024 IEEE International Conference on Communications Workshops (ICC Workshops), 2024, pp. 650–655.

[14] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," Ecography, vol. 40, no. 8, pp. 913–929, 2017.

[15] A. Apicella, F. Isgrò, and R. Prevete, "Don't push the button! Exploring data leakage risks in machine learning and transfer learning," Artificial Intelligence Review, vol. 58, no. 11, p. 339, 2025.

[16] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in Proceedings of the 33rd International Conference on Machine Learning (ICML), ser. Proceedings of Machine Learning Research, vol. 48. New York, NY, USA: PMLR, 2016, pp. 1050–1059. [Online]. Available: https://proceedings.mlr.press/v48/gal16.html

[17] X. Zhai, J. Jiang, A. Dejl, A. Rago, F. Guo, F. Toni, and A. Sivakumar, "Heterogeneous graph neural networks with post-hoc explanations for multi-modal and explainable land use inference," Information Fusion, vol. 120, p. 103057, 2025.

[18] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in International Conference on Learning Representations (ICLR), 2018. [Online]. Available: https://openreview.net/forum?id=rJXMpikCZ

[19] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?" in International Conference on Learning Representations (ICLR), 2022. [Online]. Available: https://openreview.net/forum?id=F72ximsx7C1

[20] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in Proceedings of the 34th International Conference on Machine Learning (ICML), PMLR 70, 2017, pp. 1263–1272. [Online]. Available: https://proceedings.mlr.press/v70/gilmer17a.html

[21] H. T. Friis, "A note on a simple transmission formula," Proceedings of the IRE, vol. 34, no. 5, pp. 254–256, 1946.

[22] ITU-R, "Recommendation ITU-R P.529-3: Prediction methods for the terrestrial land mobile service in the VHF and UHF bands," International Telecommunication Union, Geneva, Switzerland, Tech. Rep., 1999, withdrawn 22 Oct. 2001. [Online]. Available: https://www.itu.int/rec/R-REC-P.529-3-199910-W/en

[23] ——, "Recommendation ITU-R P.833-10: Attenuation in vegetation," International Telecommunication Union, Geneva, Switzerland, Tech. Rep., 2021. [Online]. Available: https://www.itu.int/rec/R-REC-P.833-10-202109-I/en

[24] A. Hamada and T. Kürner, "Enhancing propagation model accuracy for frequency coordination with high-resolution terrain data," in Proceedings of the 2025 International Symposium on Electromagnetic Theory (URSI EMTS 2025). URSI – International Union of Radio Science, 2025.

[25] Ç. Yapar, F. Jaensch, R. Levie, G. Kutyniok, and G. Caire, "The first pathloss radio map prediction challenge," in ICASSP 2023 – 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023, pp. 1–2.

[26] B. Feng, M. Zheng, W. Liang, and L. Zhang, "A recent survey on radio map estimation methods for wireless networks," Electronics, vol. 14, no. 8, p. 1564, 2025.

[27] X. Li, S. Zhang, H. Li, X. Li, L. Xu, H. Xu, H. Mei, G. Zhu, N. Qi, and M. Xiao, "RadioGAT: A joint model-based and data-driven framework for multi-band radiomap reconstruction via graph attention networks," IEEE Transactions on Wireless Communications, vol. 23, no. 11, pp. 17 777–17 792, 2024.

[28] A. Bufort, L. Lebocq, and S. Cathabard, "Data-driven radio propagation modeling using graph neural networks," arXiv:2501.06236, 2025.

[29] P. Zheng, X. Wang, W. Chen, R. Sun, L. Wang, Y. Yue, and N. Cheng, "RadioAR: Autoregressive modeling for accurate radio map estimation," in Companion Proceedings of the ACM Web Conference 2026 (WWW Companion '26). Dubai, United Arab Emirates: ACM, 2026, pp. 1233–1241.

[30] C. Cheng and G.-T. Zhang, "Deep learning method based on physics informed neural network with ResNet block for solving fluid flow problems," Water, vol. 13, no. 4, p. 423, 2021.

[31] J. Celades-Martínez, M. E. Diago-Mosquera, A. Peña, and J. García, "Shadow fading prediction at 18 GHz through physics guided learning in vegetative corridors," Scientific Reports, vol. 16, no. 1, p. 5916, 2026.

[32] P. L. Guth, S. Trevisani, C. H. Grohmann, J. B. Lindsay, and H. I. Reuter, "Benchmarking elevation plus land surface parameters finds FathomDEM and Copernicus DEM win as best global DEMs," Remote Sensing, vol. 17, no. 23, p. 3919, Dec. 2025. [Online]. Available: http://dx.doi.org/10.3390/rs17233919

[33] S. Trevisani and P. L. Guth, "Surface roughness in geomorphometry: From basic metrics toward a coherent framework," Remote Sensing, vol. 17, no. 23, p. 3864, Nov. 2025. [Online]. Available: http://dx.doi.org/10.3390/rs17233864

[34] G. Amatulli, D. McInerney, T. Sethi, P. Strobl, and S. Domisch, "Geomorpho90m, empirical evaluation and accuracy assessment of global high-resolution geomorphometric layers," Scientific Data, vol. 7, no. 1, p. 162, 2020.

[35] N. Mzid, S. Pignatti, W. Huang, and R. Casa, "An analysis of bare soil occurrence in arable croplands for remote sensing topsoil applications," Remote Sensing, vol. 13, no. 3, p. 474, 2021.

[36] Y. Du, Y. Zhang, F. Ling, Q. Wang, W. Li, and X. Li, "Water bodies' mapping from Sentinel-2 imagery with modified normalized difference water index at 10-m spatial resolution produced by sharpening the SWIR band," Remote Sensing, vol. 8, no. 4, p. 354, 2016.

[37] X. Zhou, B. Chen, Y. Gui, and L. Cheng, "Conformal prediction: A data perspective," ACM Computing Surveys, vol. 58, no. 2, pp. 49:1–49:37, 2025.

[38] S. H. Zargarbashi, S. Antonelli, and A. Bojchevski, "Conformal prediction sets for graph neural networks," in Proceedings of the 40th International

Conference on Machine Learning (ICML), ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 12 292–12 318. [Online]. Available: https://proceedings.mlr.press/v202/h-zargarbashi23a.html

[39] Airbus Defence and Space GmbH, "Copernicus DEM: Copernicus digital elevation model product handbook," Airbus Defence and Space GmbH, Product Handbook Version 5.0, AO/1-9422/18/I-LG, Nov. 2022. [Online]. Available: https://dataspace.copernicus.eu/sites/default/files/media/files/2024-06/geo1988-copernicusdem-spe-002_producthandbook_i5.0.pdf

[40] M. F. J. Wilson, B. O'Connell, C. Brown, J. C. Guinan, and A. J. Grehan, "Multiscale terrain analysis of multibeam bathymetry data for habitat mapping on the continental slope," Marine Geodesy, vol. 30, no. 1–2, pp. 3–35, 2007.

[41] L. Laipelt, B. Comini de Andrade, W. Collischonn, A. de Amorim Teixeira, R. C. Dias de Paiva, and A. Ruhoff, "ANADEM: A digital terrain model for South America," Remote Sensing, vol. 16, no. 13, p. 2321, 2024.

[42] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese, and P. Bargellini, "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," Remote Sensing of Environment, vol. 120, pp. 25–36, May 2012. [Online]. Available: http://dx.doi.org/10.1016/j.rse.2011.11.026

[43] R. Dubayah, J. B. Blair, S. Goetz, L. Fatoyinbo, M. Hansen, S. Healey, M. Hofton, G. Hurtt, J. Kellner, S. Luthcke, J. Armston, H. Tang, L. Duncanson, S. Hancock, P. Jantz, S. Marselis, P. L. Patterson, W. Qi, and C. Silva, "The global ecosystem dynamics investigation: High-resolution laser ranging of the Earth's forests and topography," Science of Remote Sensing, vol. 1, p. 100002, 2020.

[44] T. Markus, T. Neumann, A. Martino, W. Abdalati, K. Brunt, B. Csatho, S. Farrell, H. Fricker, A. Gardner, D. Harding, M. Jasinski, R. Kwok, L. Magruder, D. Lubin, S. Luthcke, J. Morison, R. Nelson, A. Neuenschwander, S. Palm, S. Popescu, C. K. Shum, B. E. Schutz, B. Smith, Y. Yang, and J. Zwally, "The Ice, Cloud, and land Elevation Satellite-2 (ICESat-2): Science requirements, concept, and implementation," Remote Sensing of Environment, vol. 190, pp. 260–273, 2017.

[45] A. L. Neuenschwander and L. A. Magruder, "Canopy and terrain height retrievals with ICESat-2: A first look," Remote Sensing, vol. 11, no. 14, p. 1721, 2019.

[46] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," Journal of Machine Learning Research, vol. 15, no. 56, pp. 1929–1958, 2014. [Online]. Available: https://jmlr.org/papers/v15/srivastava14a.html

[47] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in Advances in Neural Information Processing Systems, vol. 30. Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/5dd9db5e033da9c6fb5ba83c7a7ebea9-Abstract.html

[48] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778.

[49] P. J. Huber, "Robust estimation of a location parameter," The Annals of Mathematical Statistics, vol. 35, no. 1, pp. 73–101, 1964.

[50] R. K. Jaiswal, M. Elnourani, S. Deshmukh, and B. Beferull-Lozano, "A data-driven transfer learning method for indoor radio map estimation," IEEE Transactions on Vehicular Technology, vol. 75, no. 3, pp. 4261–4277, 2026.

[51] F. Wilcoxon, "Individual comparisons by ranking methods," Biometrics Bulletin, vol. 1, no. 6, pp. 80–83, 1945.

[52] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," Journal of Machine Learning Research, vol. 7, pp. 1–30, 2006. [Online]. Available: https://jmlr.org/papers/v7/demsar06a.html

[53] P. Ploton, F. Mortier, M. Réjou-Méchain, N. Barbier, N. Picard, V. Rossi, C. Dormann, G. Cornu, G. Viennois, N. Bayol, A. Lyapustin, S. Gourlet-Fleury, and R. Pélissier, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," Nature Communications, vol. 11, no. 1, p. 4540, 2020.

[54] ITU-R, "Recommendation ITU-R P.526-16: Propagation by diffraction," International Telecommunication Union, Geneva, Switzerland, Tech. Rep., Nov. 2025. [Online]. Available: https://www.itu.int/rec/R-REC-P.526-16-202511-I/en

[55] H. Meyer and E. Pebesma, "Predicting into unknown space? Estimating the area of applicability of spatial prediction models," Methods in Ecology and Evolution, vol. 12, no. 9, pp. 1620–1633, 2021.

[56] R. Valavi, J. Elith, J. J. Lahoz-Monfort, and G. Guillera-Arroita, "blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models," Methods in Ecology and Evolution, vol. 10, no. 2, pp. 225–232, 2019.

[57] V. Marsocci, N. Gonthier, A. Garioud, S. Scardapane, and C. Mallet, "GeoMultiTaskNet: remote sensing unsupervised domain adaptation using geographical coordinates," in 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2023, pp. 2075–2085.

[58] M. Guo, F. Xiong, Y. Huang, Z. Zhang, and J. Zhang, "A multi-path feature extraction and transformer feature enhancement DEM super-resolution reconstruction network," Remote Sensing, vol. 17, no. 10, p. 1737, 2025.

[59] X. Cai and M. D. Wilson, "JSPSR: Joint spatial propagation super-resolution networks for enhancement of bare-earth digital elevation models from global data," Remote Sensing, vol. 17, no. 21, p. 3591, 2025.



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